Baryons and the Bost–Connes System

Shao-Wu Hu

Shanghai, China

forest@xindong.com

ORCID: 0009-0009-8300-4691

Abstract

The survival of a baryon as a whole, its internal frequency shifts, and its transport response require a dynamical description of the same carrier. We work in the state space established by the finite p-adic reduction of the Bost–Connes system and incorporate the collective mode, three-color singlet, and flavor–spin structure into coherent formation, sustained propagation, and return. A full-spectrum reference and a common background calibration give a common mass scale of 936.669 MeV; the complete port matrix elements of a common exchange source further determine the conjugate mass-arm strengths for the stated routing. With fields and mobile compensating matter included, the implemented reference readout contains the frequency differences between two complete histories, while the arm parameters and field propagation jointly enter the memory kernel for transport. Finite inversion of coherence and continuity readouts constrains the response to a control protocol not used in parameter estimation; a pattern-preserving preparation of the complete carrier at rest and the corresponding mass spectral combination remain to be identified. Under their stated input conditions, the original phase–mass and electromagnetic self-energy prescriptions give $m_n - m_p = 1.30363$ MeV. The same internal structure yields the flavor–spin coefficients of the leading magnetic response, while weak transitions use the normalized return model of the preceding work. Nucleon endpoints, meson propagation, and return in both directions jointly produce central and tensor terms of finite range.

Contents

1 Introduction **3**

2 The Structure of Baryons **3**
2.1 The Baryon Collective Mode . . . 3
2.2 The Three-Color Singlet . . . 4
2.3 The Flavor–Spin Structure of Nucleons . . . 5

3 The Dynamics of Baryon Formation **6**
3.1 Establishing Coherence across Shells . . . 6
3.2 Sustained Propagation of the Collective Mode . . . 7
3.3 Dynamical Criteria for Stability . . . 9

4 Baryon Mass Spectra **10**
4.1 The Dynamical Origin of the Common Mass Scale . . . 10
4.2 The Neutron–Proton Mass Difference . . . 11
4.3 Phase Readout and Motion of the Complete Carrier . . . 14
4.4 Constraints on Independent Motion from Finite Readouts . . . 17
4.5 Mass Splittings within Baryon Multiplets . . . 22
4.6 Baryon Excitations . . . 24

5 Electroweak Response of Nucleons **25**
5.1 Nucleon Magnetic Moments . . . 25
5.2 Nucleon Axial Response . . . 26
5.3 Neutron Weak Decay . . . 27

6 Interactions between Nucleons **28**
6.1 Formation of Meson Exchange . . . 28
6.2 Channel Structure of the Nuclear Force . . . 30
6.3 Spatial Response of the Nuclear Force . . . 32

7 Conclusions **33**

References **35**

§1 Introduction

A baryon participates in physical processes as a whole with a definite mass, charge, and spin. Its internal structure manifests itself through electromagnetic responses, weak transitions, and exchange between nucleons. The proton and neutron share similar mass scales but differ in charge and survival properties. These similarities and differences call for an account of how a collective structure forms and persists, and how internal interactions change its phase and outputs, all within the same physical object.

The Bost–Connes system connects an arithmetic algebra with an equilibrium statistical structure exhibiting a phase transition. [1] *Dynamics of the Bost–Connes State Space* constructs the background, excitations, and their joint exchanges from its finite p-adic reduction, incorporating formation, return, restoration, and records into the complete evolution. [2] *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System* further establishes the baryon collective mode, its internal color, flavor, and spin structure, and a realization in which mass is read from a sustained interaction. [3] These results provide the common starting point of this paper.

We investigate how the sustained phase accumulation and motion response of a nucleon are connected by the same complete interaction. Background exchange preserves the collective mode; internal endpoints regulate frequency shifts and outputs; fields and compensating matter evolve together with the nucleon. The actual matrix elements of a common source and the readout of relative phase thus provide a concrete basis for relating the overall mass scale, internal differences, and motion response. Throughout, we use $\hbar = c = 1$ and express electric charge in units of the elementary charge.

§2 The Structure of Baryons

The collective mode and internal quantum numbers of a baryon play different physical roles. The collective mode describes how background components participate coherently, whereas internal color, flavor, and spin specify the transformation properties and distinguishable responses of the whole. Realizing them on a single carrier makes it possible to compare different nucleons through their subsequent dynamics.

§2.1 The Baryon Collective Mode

Following the finite p-adic reduction, take a prime p and depth $K \geq 1$. The address space is $X_K = \mathbb{Z}/p^K\mathbb{Z}$, containing $d = p^K$ addresses. The mode state space $\mathcal{H}_K$ has the orthonormal basis $\{|x\rangle : x \in X_K\}$. Shell j is $S_j = p^j X_K \setminus p^{j+1} X_K$ for $j < K$, and the terminal shell is $S_K = \{0\}$. Their normalized average states and fractional weights are

$$|e_j\rangle = \frac{1}{\sqrt{|S_j|}} \sum_{x \in S_j} |x\rangle, \qquad h_j = \frac{|S_j|}{d}, \qquad \sum_{j=0}^{K} h_j = 1. \tag{2.1}$$

The collective mode over the full address space established in the preceding work is [2, 3]

$$\boxed{|b\rangle = \frac{1}{\sqrt{d}} \sum_{x \in X_K} |x\rangle = \sum_{j=0}^{K} \sqrt{h_j}\, |e_j\rangle.} \tag{2.2}$$

The shell amplitudes combine relative to the same phase reference to form a single pure state. We write $P_\psi = |\psi\rangle\langle\psi|$ for the pure-state projector associated with a normalized vector. The corresponding density operator P_b contains the intershell matrix elements $\langle e_j|P_b|e_\ell\rangle = \sqrt{h_j h_\ell}$. The shell-dephased background $\sum_j h_j P_{e_j}$ has the same shell weights but lacks these off-diagonal terms. Formation of this collective mode therefore requires both the participation weights and coherence across shells to be established.

For the multiplicative pullbacks $(T_n f)(x) = f(nx)$, the preceding work proves that the common fixed space is

$$T_n b = b \quad (n \geq 1), \qquad \bigcap_{n \geq 1} \ker(T_n - I) = \operatorname{span}\{b\}. \tag{2.3}$$

This common property involves multiplication across shells. Invariance under the unit group alone still permits independent constants on different shells, giving the space $\mathcal{C} = \operatorname{span}\{e_0, \ldots, e_K\}$. The unit-shell average state e_0 and collective mode b therefore represent different participation structures. A multiplicative pullback is generally nonunitary when $p \mid n$. Equation (2.3) specifies the arithmetic property of the mode; its physical preservation is realized by the self-adjoint interaction in Chapter 3.

The address x labels a mode component; propagation position is represented by a separately realized positional degree of freedom. A common amplitude over all addresses does not, by itself, specify the spatial size of a nucleon. Internal color, flavor, and spin are likewise described in their respective representation spaces. Using the matter carrier $\mathcal{H}_K \otimes \mathcal{H}_{\text{int}}$ of the preceding work, a normalized joint state whose mode marginal is the pure state P_b has the form [3]

$$\Omega_B = P_b \otimes \sigma_B, \qquad \sigma_B \geq 0, \qquad \operatorname{Tr} \sigma_B = 1. \tag{2.4}$$

Here σ_B may contain internal entanglement or mixing. Different internal preparations can share the same collective mode and hence a common mode response, while differing in charge, spin, and internal exchange. The internal color structure first determines which components can form a color singlet.

§2.2 The Three-Color Singlet

In the $p = 7$ matter correspondence adopted in the preceding work, the fundamental color carrier is $V_c = \mathbb{C}^3$, and color transformations act through the fundamental representation of $SU(3)$. [3] The normalized singlet of three fundamental color factors is

$$|\varepsilon_3\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c=1}^{3} \varepsilon^{abc} |a, b, c\rangle, \qquad U^{\otimes 3}|\varepsilon_3\rangle = |\varepsilon_3\rangle \quad (U \in SU(3)). \tag{2.5}$$

The completely antisymmetric tensor contains each of the three color labels once, and the whole is invariant under a change of color reference. The representation-theoretic result established previously shows that, among nonempty tensor products consisting solely of fundamental representations, a singlet first occurs in the triple tensor product, whose singlet subspace is spanned by ε_3. Combinations of fundamental and conjugate representations have a different singlet structure; meson exchange will be treated below using its actual carrier.

The three tensor slots in Eq. (2.5) label internal color factors, not three propagation positions. One factor of each color does not require three distinct flavor labels, so the color singlet can accompany the proton combination uud or neutron combination udd. The color part is antisymmetric under exchange, and the flavor, spin, and internal spatial parts must be organized accordingly to form a complete nucleon state.

The reduced state at any color slot is $I_3/3$. This internal mixed state reflects correlations in the pure color singlet as a whole; local color readouts are specified by the corresponding internal operators. Physical separation also involves an actual interaction and the survival of the separated carrier. Chapter 3 treats it through a process criterion for colored outputs. The charge and spin of the nucleon are determined by the remaining factors of the same internal state.

§2.3 The Flavor–Spin Structure of Nucleons

In the nucleon representation of the preceding work, the flavor space is spanned by u, d, with charges $2/3, -1/3$, respectively. Spin carries the two-dimensional representation of $SU(2)$, with basis $\uparrow, \downarrow$. These flavor, charge, and nucleon-composition labels agree with the quark classification. [4] The preceding work realizes the corresponding representations and exchange rules on an actual internal carrier, while the collective mode remains invariant under permutations of internal slots. [3]

To fix the notation for subsequent matrix elements, we retain the two sets of permutation vectors used in the preceding work. The normalized spin vectors with projection $+1/2$ are

$$\eta_\rho = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}}, \qquad \eta_\lambda = \frac{2|\uparrow\uparrow\downarrow\rangle - |\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{6}}. \tag{2.6}$$

The flavor vectors have the same permutation form:

$$\begin{aligned} \phi_{p,\rho} &= \frac{|udu\rangle - |duu\rangle}{\sqrt{2}}, \\ \phi_{p,\lambda} &= \frac{2|uud\rangle - |udu\rangle - |duu\rangle}{\sqrt{6}}, \\ \phi_{n,a} &= \phi_{p,a}\big|_{u\leftrightarrow d} \quad (a = \rho, \lambda). \end{aligned} \tag{2.7}$$

The subscripts ρ, λ merely distinguish the two permutation vectors. The corresponding flavor–spin state is

$$\Phi_{N,\uparrow} = \frac{\phi_{N,\rho} \otimes \eta_\rho + \phi_{N,\lambda} \otimes \eta_\lambda}{\sqrt{2}}, \qquad N = p, n. \tag{2.8}$$

The preceding work verifies the normalization of this combination, its total spin $1/2$, and its symmetry under simultaneous permutations of flavor and spin slots. [3]

For a symmetric internal spatial state, the mode, color, and flavor–spin parts of the nucleon are

$$\boxed{|B_{N,\uparrow}\rangle = |b\rangle \otimes |\varepsilon_3\rangle \otimes \Phi_{N,\uparrow}, \qquad N = p, n.} \tag{2.9}$$

The common symmetric spatial factor is suppressed here. The antisymmetric color part and symmetric flavor–spin part together satisfy the adopted fermionic exchange rule. The statistical problem posed by spin–flavor symmetry was discussed in early work; [5] here it is accommodated by the explicit color singlet of the preceding work. If the permutation type of the internal spatial state changes, the other factors must retain the corresponding exchange properties.

Let $q = \mathrm{diag}(2/3, -1/3)$ denote the charge operator on one flavor factor. The total charge Q_{em} is the sum of q over the three slots and acts as the identity on the color, spin, and mode factors. Since every component of Eq. (2.7) has fixed flavor numbers, the nucleon states give

$$Q_{\mathrm{em}}\Phi_{p,\uparrow} = \Phi_{p,\uparrow}, \qquad Q_{\mathrm{em}}\Phi_{n,\uparrow} = 0. \tag{2.10}$$

The proton and neutron share b and ε_3 and are distinguished by Φ_N. Formation of this complete nucleon state requires coherence across the shells of the collective mode and preservation of its internal structure during subsequent propagation.

§3 The Dynamics of Baryon Formation

Formation of the collective mode requires definite relative phases across shells; persistence requires an actual interaction that preserves the mode and its internal structure. Exchange, retrieval, and restoration in the mathematical framework provide joint realizations of these processes. [2, 3] The mode, internal state, reference, and auxiliaries after use all participate in this evolution, and the formed nucleon continues to propagate along the same joint history.

§3.1 Establishing Coherence across Shells

Preparations with identical shell weights may have different coherences. Let $\mathcal{E}$ be the Hilbert space carrying the internal state and all other participating degrees of freedom. A normalized joint pure state supported on $\mathcal{C} \otimes \mathcal{E}$ can be written as $\Psi = \sum_j e_j \otimes \xi_j$. Its mode matrix elements are determined by overlaps of the remaining components:

$$\begin{gathered} \rho_{\mathrm{mode}} = \sum_{j,\ell} \langle \xi_\ell | \xi_j \rangle |e_j\rangle\langle e_\ell|, \\ \rho_{\mathrm{mode}} = P_b \quad \Longleftrightarrow \quad \xi_j = \sqrt{h_j}\,\xi \text{ for all } j, \quad \|\xi\| = 1. \end{gathered} \tag{3.1}$$

This is the common-final-state criterion of the preceding work. [3] In the formation output, every shell must connect to the same state of the remaining degrees of freedom in order to preserve the intershell coherence of Eq. (2.2). The internal state may still be correlated with other auxiliaries; these correlations are contained in ξ. Background restoration in Ref. [2] produces a shell mixture with specified weights.

The pure mode P_b is instead obtained by exchange involving a coherent reference. [3]

Take an isospectral reference copy $\mathcal{C}_R$ of the shell-average space $\mathcal{C}$. The bare mode energy obeys $H_K e_j = E_j e_j$; choose the reference Hamiltonian $H_R e_j^R = E_j e_j^R$ and prepare $b_R = \sum_j \sqrt{h_j} e_j^R$. The exchange given in the preceding work acts on the two shell-average spaces, with the remaining directions acting as spectators:

$$\begin{aligned} U_R(e_j \otimes e_k^R) &= e_k \otimes e_j^R, \\ [U_R, H_K + H_R] &= 0, \\ U_R(c \otimes b_R) &= b \otimes c_R. \end{aligned} \tag{3.2}$$

Here $c \in \mathcal{C}$, and c_R is its copy in the reference space. The two exchange endpoints have the same bare energy spectrum. Both the shell-average states and the reference carry trivial unit-group charge, so the joint exchange accounts for the corresponding energy and group charge. For an arbitrary joint input supported on $\mathcal{C} \otimes \mathcal{E}$, the complete successor state is

$$(U_R \otimes I_\mathcal{E})\big(\rho_{\mathcal{C}\mathcal{E}} \otimes P_{b_R}\big)(U_R^\dagger \otimes I_\mathcal{E}) = P_b \otimes \rho_{R\mathcal{E}}. \tag{3.3}$$

The tensor factors are ordered according to the exchange. The state $\rho_{R\mathcal{E}}$ is the same original joint state with its mode factor transferred to the reference. It retains the original correlations; after tracing out the reference, the internal marginal remains $\mathrm{Tr}_\mathcal{C} \rho_{\mathcal{C}\mathcal{E}}$. When the internal preparation consists of the color singlet and nucleon flavor–spin state in Chapter 2, the exchange establishes the required b while preserving the internal nucleon structure.

Subsequent use of the reference starts from the joint state $\rho_{R\mathcal{E}}$ left by Eq. (3.3). Bare free evolution also gives different phases to different shells. The formed nucleon must therefore be coupled to a sustained interaction that preserves the complete mode.

§3.2 Sustained Propagation of the Collective Mode

The sustained interaction preserves the mode as b on both the outward and return paths, while the interaction port accumulates a readable phase. Following the finite spectral-path realization of the preceding work, take a positive integer spectral cutoff $L \geq 2$ and weight parameter $\beta > 1$, and set $w_n = n^{-\beta}$ and $Z_L = \sum_{n=1}^{L} w_n$. The spectral label n and finite address depth K count different participants.

Retain orthogonal auxiliary ports $|z\rangle_P, |1\rangle_P, \ldots, |L\rangle_P$ outside the mode factor, where z is a port label. Let $J_b\psi = b \otimes \psi$, with ψ containing these ports and the complete internal state. Pairing each multiplicative path with its port, the self-adjoint interaction and simultaneous equal-strength protection of its two arms are [3]

$$\begin{aligned} A_B &= (\log Z_L) \sum_{n=1}^{L} \sqrt{w_n}\big(T_n \otimes |n\rangle_P\langle z| + T_n^\dagger \otimes |z\rangle_P\langle n|\big), \\ S_b &= 2J_bJ_b^\dagger - I, \qquad \widehat{A}_B = \tfrac{1}{2}(A_B + S_bA_BS_b). \end{aligned} \tag{3.4}$$

The identity on internal factors is suppressed. From $T_n b = b$, the compression within the mode is $J_b^\dagger A_B J_b = X_B$. Since $S_b J_b = J_b$, summing the two arms cancels the components outside the mode and yields the full intertwining relation: [3]

$$\begin{aligned} \widehat{A}_B J_b &= \frac{I + S_b}{2} A_B J_b = J_b J_b^\dagger A_B J_b = J_b X_B, \\ X_B &= |a_B\rangle\langle z|_P + |z\rangle_P\langle a_B|, \\ |a_B\rangle &= (\log Z_L) \sum_{n=1}^{L} \sqrt{w_n} |n\rangle_P. \end{aligned} \tag{3.5}$$

This relation ensures that the preserved subspace specified by J_b is invariant under the interaction. The port evolves under X_B, while the collective mode and internal color state are retained. The coherent paths and their reverse actions thus constitute a complete sustained exchange. The nonunitary part of the original multiplicative pullback enters as a matrix element of a self-adjoint interaction, and energy compensation and control resources are included through the joint realization of the preceding work.

Orthogonality of the ports gives $\|a_B\|^2 = Z_L(\log Z_L)^2$. The two bright states of the port interaction and their eigenvalues are therefore

$$|\pm_B\rangle = \frac{|z\rangle_P \pm |a_B\rangle/r_B}{\sqrt{2}}, \qquad X_B|\pm_B\rangle = \pm r_B|\pm_B\rangle, \qquad r_B = \sqrt{Z_L}\log Z_L. \tag{3.6}$$

After formation, we use the positive-bright path-port preparation $|+_B\rangle$ of Ref. [3] and couple it to the established interaction for motion. This preparation fixes the path port, while retaining orientation and its correlations with all remaining degrees of freedom. The negative bright branch and dark branches remain in the complete space. Separately, let Y_A denote the mass-orientation operator and Z_A the anticommuting transport-orientation operator, with $Y_A^2 = Z_A^2 = I$. The complete mass term is $g\widehat{A}_B \otimes Y_A$, which becomes $gr_B Y_A$ on the positive bright port. The bright-state sign of the port and this pair of orientations belong to different factors. Along one positional direction, during a constant active interval, the propagation generator on the preserved positive-bright subspace is

$$H_B(k) = gr_B Y_A + 2\omega Z_A \sin(k\ell_x), \qquad E_\pm(k) = \pm\sqrt{g^2 r_B^2 + 4\omega^2 \sin^2(k\ell_x)}. \tag{3.7}$$

Here $g > 0$ is the collective interaction strength, ω the position-exchange strength, ℓ_x the length calibration between adjacent positions, and k the propagation wave number. Translation preserves the mode, path port, and internal color singlet, so the same nucleon carrier persists in motion. In this positive-bright preservation model for a fixed active interval, the two orientation branches are symmetric about the established zero center, and the half-gap at zero wave number is gr_B. This coefficient also enters the dispersion relation of the generator; an absolute readout additionally requires the full elapsed time and a common background calibration. Fields and mobile compensating matter that do not commute with the transport edge change both the complete generator and the readable phase. Chapter 4 retains these degrees of freedom to examine how the original mass arm continues to participate in motion.

Complete propagation also includes thermal processing, temporary storage, and return. The isospectral storage of the preceding work preserves the entire mode, port, and their internal correlations. Upon success, the inverse exchange returns the original carrier; failure branches are retained together with the stored state and actual outputs. [3] In this realization, the active interaction is switched off during processing, and the bare free phases of the system and storage copy are physically compensated. Switching work and compensation controls are included in the complete process. The time required for initial formation and port preparation enters the total history according to the actual number of occurrences. The repeated active interaction, storage, thermal processing, control, waiting, and return all enter the elapsed time of each cycle. Dividing the cycle phase by this full elapsed time gives the frequency readout for the same process.

§3.3 Dynamical Criteria for Stability

The sustained interaction preserves the nucleon mode and internal structure, while return and exit channels determine the survival share. Begin with a nucleon at rest and a fixed environment preparation. Let $|\mathcal{B}\rangle$ denote the complete carrier state containing the mode, internal state, and selected interaction ports. Write R_r for the return operators and D_s for the exit operators of one complete process, whose full set of outcomes satisfies $\sum_r R_r^\dagger R_r + \sum_s D_s^\dagger D_s = I$. Under the survival criterion of the preceding work, return that preserves the same carrier obeys

$$R_r|\mathcal{B}\rangle = c_r|\mathcal{B}\rangle, \qquad s_B = \sum_r |c_r|^2. \tag{3.8}$$

The squared moduli of the return coefficients give the survival share. Phase readouts are determined by cross coherences between return branches and a common reference, combined according to the actual record structure. If $D_s|\mathcal{B}\rangle = 0$ for every exit in the complete outcome set, then $s_B = 1$. A nonzero exit probability for the input instead gives $s_B < 1$ and reduces the survival of the nucleon in that process. [3] Subsequent processes must retain correlations with the environment and records left after use, and compute continued survival along the actual joint successor states. The collective mode or lowest collective scale alone does not replace an examination of exit channels.

Color separation is also assessed from actual outputs. Fix the input-support projector Π_S and the output object to be separated. Let Q_{col} project onto the nonsinglet color representations of that object, and let K_r be the complete implemented outcome operators. The criterion for no colored output is [3]

$$\boxed{Q_{\mathrm{col}} K_r \Pi_S = 0 \qquad \text{for every actual outcome } r.} \tag{3.9}$$

This condition holds when the input is a color singlet and every outcome operator intertwines the input representation with the color representation of the selected output object. If the environment can carry compensating color, total color conservation must still be assessed together with the representation of the output object and its subsequent preservation. The internal color partial trace in Chapter 2 describes only local correlations. Independent colored matter requires a separation interaction and an output carrier capable of persisting; Eq. (3.9) accordingly constrains color separation in the process under consideration.

The proton and neutron share a collective mode, while internal transitions can change their return phases and exit shares. The former enter the mass readings of the next chapter; the latter determine survival through actual transitions and their subsequent outputs.

§4 Baryon Mass Spectra

The common nucleon mode sets the phase scale of the sustained interaction, while internal flavor, spin, and exchange structures modify its frequency shifts. The preceding work computes this scale and the nucleon mass difference using a specified spectral preparation and common background calibration. [3] Connecting these readings to the motion of the complete nucleon additionally requires identifying the exchange source that supplies the actual strength of the internal frequency shift, and the pattern-preserving carrier that still supports a rest-mass readout once fields and compensating matter are included. We retain the original evaluation prescriptions and develop these two dynamical connections within the same complete interaction.

§4.1 The Dynamical Origin of the Common Mass Scale

After nucleon formation and preparation of the positive bright port, the positive-orientation rest branch accumulates phase at the rate gr_B during the active interval. Consider a surviving history h in which the mode and port return completely. Let T_h be the total duration of active interaction, processing, waiting, and return, and set $g(t) = 0$ during intervals in which the active interaction is switched off. With the bare-phase compensation described in Chapter 3, return preserves the same phase readout. The average drive over the complete history and the corresponding mass are

$$\bar{g}_h = \frac{1}{T_h} \int_h g(t)\, dt, \qquad m_{B,h} = \bar{g}_h r_B. \tag{4.1}$$

For this pattern-preserving branch at rest, Eq. (4.1) reads frequency over the full elapsed duration. When using the beat frequency of two branches, the actual common reference must establish the center c_h and the branches must be verified to be $c_h \pm m_{B,h}$ before the half-gap can yield the same mass reading. The time and resources for initial formation and port preparation are accounted for separately as in §3.2; a general state during those stages does not accumulate phase according to a fixed bright-state eigenvalue.

The absolute energy unit is calibrated jointly against the actual background. Using the background port of the preceding work, [3] write $r_{\mathrm{vac}} > 0$ for its dimensionless bright-state eigenvalue, b_r for the squared weights of the gradient branches, and C_{bg} for their sum. The background and nucleon share the drive, active window, and final readout time, so the full-cycle frequency of each background branch is $\Omega_{r,h} = \bar{g}_h r_{\mathrm{vac}} \sqrt{b_r}$. Matching the squared responses using the energy scale E_0 gives

$$\frac{\sum_r \Omega_{r,h}^2}{E_0^2} = \frac{\bar{g}_h^2 r_{\mathrm{vac}}^2 C_{\mathrm{bg}}}{E_0^2} = r_{\mathrm{vac}}^2 \quad \Longrightarrow \quad \bar{g}_h = \frac{E_0}{\sqrt{C_{\mathrm{bg}}}}. \tag{4.2}$$

The common background bright-state factor cancels in this matching, leaving the average drive including pauses and processing time. Denoting the nucleon spectral parameter by β_B and substituting Eq. (3.6), we obtain

$$\boxed{M_B(L,\beta_B) = E_0\sqrt{\frac{Z_L(\beta_B)}{C_{\mathrm{bg}}}}\log Z_L(\beta_B).} \tag{4.3}$$

The shared phase of the spectral paths supplies the factor $\sqrt{Z_L}$ in the port eigenvalue, the weight of the sustained interaction supplies $\log Z_L$, and background calibration converts this dimensionless response into a mass. All three refer to the same preparation, interaction, and duration, allowing comparison with the other matter readings of the preceding work.

The spectral parameter follows the leading-variance prescription of the preceding work. For the normalized full-spectrum weights $n^{-\beta}/\zeta(\beta)$, the variance of $\log n$ is $V(\beta) = \partial_\beta^2 \log\zeta(\beta)$, with leading term $V_{\mathrm{lead}}(\beta) = (\beta-1)^{-2}$ as $\beta \downarrow 1$. The nucleon spectral preparation sets $V_{\mathrm{lead}}(\beta_B) = 3$, hence $\beta_B = 1 + 1/\sqrt{3}$. In this full-spectrum reference, the background response is $C_{\mathrm{bg}} = C^*$, and we use

$$\begin{aligned} Z_\infty(\beta_B) &= \zeta(\beta_B), \\ C^* &= 2\sum_{n=1}^{\infty}\log^2\!\left(1+\frac{1}{n}\right) = 1.9543783665\ldots, \\ M_B &= 936.669312626\ldots\ \text{MeV} \qquad (E_0 = 1\ \text{GeV}). \end{aligned} \tag{4.4}$$

Equation (4.4) uses this full-spectrum reference. A finite realization instead uses its own spectral parameter, cutoff, and background response in Eq. (4.3). The proton and neutron share the mass baseline of the same spectral preparation, on which internal frequency shifts are subsequently superposed.

§4.2 The Neutron–Proton Mass Difference

The two nucleons in Chapter 2 have identical mode and color parts, with flavor compositions *uud* and *udd*, respectively. Following the character convention on the unit group modulo 7 used in the preceding work, set $\chi_1(3) = e^{i\pi/3}$ and $\chi_5 = \overline{\chi_1}$ and extend both characters by zero on multiples of 7. For $s > 1$, the corresponding Dirichlet L-function is $L(s,\chi) = \sum_{n\geq 1}\chi(n)n^{-s}$. Under the flavor–direction correspondence of the preceding work, [3] u, d use the spectral responses of χ_1, χ_5, respectively. At the nucleon spectral point, take $L_1 = L(\beta_B, \chi_1)$ and $L_5 = \overline{L_1}$. With flavor projectors P_u, P_d, define the single-slot flavor response $F_L = L_1 P_u + L_5 P_d$, acting as the identity on spin. Every proton permutation in Chapter 2 contains two copies of u and one of d, while the neutron has the reverse composition. The coherent product response therefore satisfies

$$\begin{aligned} F_L^{\otimes 3}\Phi_N &= a_N\Phi_N, \quad N = p, n, \\ a_p &= L_1^2 L_5 = |L_1|^2 L_1, & a_n &= \overline{a_p}, \\ \Theta_p &= \theta, & \Theta_n &= -\theta, \\ \theta &= \arg L_1 = 0.330191432349\ldots, & \tan\theta &= 0.342738773379\ldots. \end{aligned} \tag{4.5}$$

Here arg is the principal argument. Every permutation in the flavor–spin state retains the same total phase. Converting it into an actual frequency shift requires an exchange acting on the complete spectral auxiliary. Take the weights $c_j = \sum_{m\ge 0}(7m+j)^{-\beta_B}$ of the six unit residue classes, capacity $C_{\beta_B} = \sum_{j=1}^{6} c_j = (1-7^{-\beta_B})\zeta(\beta_B)$, and actual spectral preparation $|r_{\beta_B}\rangle = \sum_j \sqrt{c_j/C_{\beta_B}}|j\rangle$. Following the original controlled group action, [3] use $D_j^{\mathrm{fl}} = \chi_1(j)P_u + \chi_5(j)P_d$ on the flavor factor and set

$$\begin{aligned} U_\beta &= \sum_{j=1}^{6} D_j^{\mathrm{fl}} \otimes |j\rangle\langle j|, \qquad & U^{(3)} &= U_\beta^{\otimes 3}, \\ K_{\mathrm{sp}} &= \langle r_{\beta_B}^{\otimes 3}|U^{(3)}|r_{\beta_B}^{\otimes 3}\rangle = \frac{F_L^{\otimes 3}}{C_{\beta_B}^{3}}, & & \\ K_{\mathrm{sp}}\Phi_N &= (k_R + it_N k_I)\Phi_N, \qquad & t_p = 1, \quad t_n &= -1, \quad \frac{k_I}{k_R} = \tan\theta. \end{aligned} \tag{4.6}$$

The unitary action retains all outputs of the three spectral factors. The expression above is only its return matrix element, without renormalization by the return success probability. Realizing the full-spectrum reference weights in a finite auxiliary still incurs supply and precision costs. At the spectral point used here, $k_R = 0.0654205666626$ and $k_I = 0.0224221647717$; these differ from the unnormalized a_N.

We construct an explicit common routing using the complete source preparation, collective exchange, and simultaneous protection of the preceding work. The separately supplied source and target nucleon each retain their matter, work, and record degrees of freedom. Write the prepared positive-bright source column as

$$\begin{aligned} J_s &= \frac{|\mathrm{in}\rangle W_s + |\mathrm{out}\rangle V_s}{\sqrt{2}}, \quad W_s^\dagger W_s = V_s^\dagger V_s = I, \\ J_s^\dagger \Pi_{\mathrm{in}} J_s &= J_s^\dagger \Pi_{\mathrm{out}} J_s = \tfrac{1}{2} I, \qquad H_{\mathrm{src}} J_s = J_s H_L. \end{aligned} \tag{4.7}$$

The maps W_s, V_s are the complete input and output isometries of the actual loading process, including work, matching, and previous records; H_L is the interaction retained on the source input subspace. Equal port weights form an operator relation valid for every input in that subspace, allowing correlations between the source and other inputs. We condition here on actual successful supply of the positive-bright subspace, with success probability p_{act}. Supply failures, waiting, and the source after use are all retained. The complete preparation and its inverse must also be physically available to implement the joint protection below.

Let H_{par} be the complete parent exchange of Chapter 3. The map J_0 includes the target positive bright port, the preparation of the three spectral factors, and the remaining inputs, such that the parent arm gives $\mu_{\mathrm{parent}} Y_A$ in this subspace. The mass orientation Y_A and transport orientation Z_A are those of Chapter 3. The common real-part reference has the additional operator X_R, with $X_R^2 = I$. Reroute the same parent exchange at the source output and take

$$\begin{aligned} C_\alpha &= \sin\alpha X_R + i\cos\alpha I, \qquad & V_\alpha &= P_z + P_{\mathrm{out}} U^{(3)} C_\alpha, \\ D_{\mathrm{route}} &= \Pi_{\mathrm{in}} \otimes I + \Pi_{\mathrm{out}} \otimes V_\alpha, \quad & H_{\mathrm{route}} &= D_{\mathrm{route}}(I \otimes H_{\mathrm{par}}) D_{\mathrm{route}}^\dagger. \end{aligned} \tag{4.8}$$

The parameter α is the stated connection angle. Within the target parent mass port, $P_z = |z\rangle_P\langle z|$ and $P_{\mathrm{out}} = I - P_z$ are distinct from the source port Π_{out}. Suppose a forward edge of the parent arm has amplitude $A_n y$, where $y = \pm 1$ is an eigenvalue of Y_A, and the corresponding spectral-unitary eigenvalue is u_ℓ. Its reference matrix element at the source output is then $A_n y\, u_\ell (i \cos\alpha\, \delta_{z'_R z_R} + \sin\alpha\, \delta_{z'_R, 1-z_R})$, where $z_R, z'_R = 0, 1$ label the Z_R basis of the reference. The sum of the squared amplitudes of the two orthogonal outputs remains $|A_n y|^2$, and each reverse edge is the adjoint of its forward edge. The two arms thus share the same parent matrix element, without a separately assigned second gain. This realization requires these exchange edges to remain physically active simultaneously; the ability to execute routing and inverse routing sequentially is insufficient.

The original interaction H_{ref} that maintains the real-part reference, retained in full, yields λX_R on the range of J_0, with independently retained strength λ. Apply simultaneous protection to the complete prepared subspace, including the source:

$$\begin{aligned} H_{\mathrm{raw}} &= H_{\mathrm{src}} \otimes I + H_{\mathrm{route}} + I \otimes H_{\mathrm{ref}}, \\ \mathcal{J} &= J_s \otimes J_0, \qquad S = 2\mathcal{J}\mathcal{J}^\dagger - I, \\ \widehat{H} &= \tfrac{1}{2}(H_{\mathrm{raw}} + S H_{\mathrm{raw}} S), \\ \widehat{H}\mathcal{J} &= \mathcal{J}(\mathcal{J}^\dagger H_{\mathrm{raw}} \mathcal{J}) = \mathcal{J}[H_L \otimes I + I \otimes h_N]. \end{aligned} \tag{4.9}$$

The last line follows from $S\mathcal{J} = \mathcal{J}$, and the complementary subspace of the protection remains within the complete interaction. Using the half-weight of each source port and $\mathrm{Re}[(k_R + i t_N k_I) C_\alpha] = k_R \sin\alpha X_R - t_N k_I \cos\alpha I$ gives directly

$$\boxed{\begin{aligned} h_N &= (\mu_0 - t_N b_{\mathrm{arm}}) Y_A + a_{\mathrm{arm}} Y_A X_R + \lambda X_R, \\ \mu_0 &= \frac{\mu_{\mathrm{parent}}}{2}, \qquad q_{\mathrm{src}} = 1, \\ a_{\mathrm{arm}} &= \mu_0 k_R \sin\alpha, \\ b_{\mathrm{arm}} &= \mu_0 k_I \cos\alpha, \\ \chi_{\mathrm{arm}} &= \frac{a_{\mathrm{arm}}}{\mu_0} = k_R \sin\alpha. \end{aligned}} \tag{4.10}$$

The quantity q_{src} is the strength ratio of the two source-routed arms relative to the same parent exchange. The factor $1/2$ in μ_0 comes from the actual source-port matrix element; the independent reference λX_R is not halved with it. The source continues to evolve under H_L throughout the process. This intertwining neither replaces the source by an average nor restores it to its initial state. Source protection, backreaction, and control costs remain part of the complete interaction.

In a common $X_R = r = \pm 1$ branch, the active arm coefficient is $m^{\mathrm{arm}}_{N,r} = \mu_0 - t_N b_{\mathrm{arm}} + r a_{\mathrm{arm}}$, giving

$$\boxed{\Delta m_{\mathrm{arm}} = 2 b_{\mathrm{arm}}, \qquad \frac{\Delta m_{\mathrm{arm}}}{\mu_0} = 2\chi_{\mathrm{arm}} \cot\alpha\, \frac{k_I}{k_R}.} \tag{4.11}$$

Here $k_R \sin\alpha \neq 0$ is assumed; degenerate connections use the first equality directly. All reference branches and cross coherences still participate in the actual readout. The quantity μ_0 is the active baseline of the new connection. To connect it to the common mass scale in Eq. (4.4) and the original phase–mass prescription, one

must independently verify $\chi_{\text{arm}} \cot \alpha = M_B^{\text{actual}}/v_{\text{EW}}$ using the same units, preparation, and full elapsed duration. The common-source result $q_{\text{src}} = 1$ determines the ratio of the two strengths, but does not determine this calibration relation.

The contribution of electromagnetic interactions to mass splittings within baryon multiplets was analyzed in early work based on flavor symmetry. [6] Here we retain the specific numerical prescription of the preceding work: set $\delta_N^{(L)} = -(M_B^2/v_{\text{EW}}) \tan \Theta_N$, where $v_{\text{EW}} = 246.191373933\ldots$ GeV is the electroweak energy scale from the common background calibration, distinct from the transport strength. The electromagnetic coupling uses the full response $\alpha_{\text{EM}}^{-1} = 137.0363590341\ldots$. Applying the original positive-self-energy calibration and the deep-level six-path share 1/6, assign $\alpha_{\text{EM}} M_B/6$ to the proton branch and denote the remaining common frequency shift by δ_0. This gives [3]

$$\begin{aligned} m_p &= M_B + \delta_0 - \frac{M_B^2}{v_{\text{EW}}} \tan\theta + \frac{\alpha_{\text{EM}} M_B}{6}, \\ m_n &= M_B + \delta_0 + \frac{M_B^2}{v_{\text{EW}}} \tan\theta. \end{aligned} \tag{4.12}$$

Let P_p, P_n be the proton and neutron projectors in the four-state nucleon space. The effective rest model using these inputs is $H_{\text{rest}}^{(+)} = m_p P_p + m_n P_n$, and both frequency-shift terms preserve charge. These are the phase–mass and self-energy evaluation prescriptions of Ref. [3]. The positive stored energy of the complete field already has a dynamical origin, but its absolute contribution to the rest readout of a mobile nucleon must still be established. The six-path address share alone does not determine a charge distribution in propagation space and therefore cannot, by itself, identify field energy as a mass correction of the complete nucleon.

The common shift cancels upon subtraction, yielding the mass difference

$$\boxed{\begin{aligned} \Delta m := m_n - m_p &= \frac{2M_B^2}{v_{\text{EW}}} \tan\theta - \frac{\alpha_{\text{EM}} M_B}{6} \\ &= (2.442828542637 - 1.139198043531)\ \text{MeV} \\ &= 1.303630499106\ldots\ \text{MeV}. \end{aligned}} \tag{4.13}$$

In this prescription, the conjugate spectral phases raise the relative frequency of the neutron branch, while the positive electromagnetic self-energy of the proton reduces the difference. The common shift remains in both absolute masses. The subsequent weak-decay calculation inherits these inputs.

§4.3 Phase Readout and Motion of the Complete Carrier

Propagation of a charged nucleon changes both the field and the compensating matter. Using the charged transport and field exchange of the preceding work, [3, 7] retain the nucleon position shift and its corresponding flux displacement on every directed edge. Denote their combined transport operator by $S_{A,N}$ and set $D_{A,N} = (S_{A,N}^\dagger - S_{A,N})/(2i)$. A proton edge includes the flux displacement required by positive charge; the supplied negatively charged electron uses the opposite displacement. The neutral nucleon starts from the electron vacuum. Each process evolves within its own Gauss constraint and matter identity. Cancellation of certain responses in the

neutral-vacuum sector of this finite model does not imply that neutrons in general lack an internal electromagnetic response.

Extend the isometry of the preceding section by the identity on position, field, electron, and auxiliary factors. Carry the same nucleon transport on every mass and source port, rather than activating transport only at the mass output. The prepared-subspace projector then commutes with $Z_A D_{A,N}$ and the remaining complete interaction $B_N(u)$, so the original intertwining extends to

$$\begin{aligned}\widehat{H}_N(u)\mathcal{J} = \mathcal{J}\Big[H_L + B_N(u) \\ + u^2\big\{[(\mu_0 - t_N b_{\mathrm{arm}})I + a_{\mathrm{arm}} X_R]Y_A \\ + G_N Z_A D_{A,N}\big\}\Big].\end{aligned} \tag{4.14}$$

The envelope u follows the actual loading, holding, and switch-off schedule; G_N is the transport-edge strength. The operator $B_N(u)$ retains the field, active auxiliaries, independent reference, and, in the proton realization, the complete mobile electron. The source interaction H_L remains present. Below, we write only the field and nucleon interactions in the matter representation supplied by this exact intertwining, which does not require the initial state to factorize between source and matter. On the plateau $u = \delta$ and in a fixed reference branch $X_R = r$,

$$\begin{aligned} H_{N,r} &= B_{N,r} + K_{N,r}, \\ K_{N,r} &= M_{N,r} Y_A + g_{\mathrm{tr}} Z_A D_{A,N}, \\ M_{N,r} &= \delta^2(\mu_0 - t_N b_{\mathrm{arm}} + r a_{\mathrm{arm}}), \\ g_{\mathrm{tr}} &= \delta^2 G_N. \end{aligned} \tag{4.15}$$

The quantity $M_{N,r}$ is the mass-arm coefficient obtained from the same underlying joint interaction. The complete $B_{N,r}$ generally does not commute with $D_{A,N}$, so the positive and negative branches of $K_{N,r}$ are not yet the rest-mass spectrum of $H_{N,r}$. We temporarily suppress N, r below; all operators still act on the same complete sector.

The implemented reference measurement compares two finite histories. Let Ξ be the input column produced by complete loading and holding, together with previous records. The switch-off operation U_d is implemented over duration τ by time-ordered evolution under $H - sK$, with s rising from 0 to 1. The two arms either switch off before propagation or propagate before switching off, so

$$\begin{aligned} \Phi_0(t) &= e^{-iB(t+\tau)} U_d \Xi, \\ \Phi_1(t) &= U_d e^{-iH(t+\tau)} \Xi, \\ u_0 &= U_d \Xi, \qquad v_0 = e^{-iH\tau}\Xi, \\ F &= e^{iB\tau} U_d, \\ C(t) &= u_0^\dagger e^{iBt} F e^{-iHt} v_0. \end{aligned} \tag{4.16}$$

Interference converts accumulated phase into readings, as in the coherent-readout principle of Ramsey's separated-oscillating-field method. [8] Here, two orthogonal settings of an independent reference pointer read the real and imaginary parts of $C(t)$; the settings are implemented in separately supplied complete trials. The field and compensating matter are not removed from the reference history. If $H|E_a\rangle = E_a|E_a\rangle$ and $B|\epsilon_\beta\rangle = \epsilon_\beta|\epsilon_\beta\rangle$, the implemented coherence is

$$C(t) = \sum_{\beta,a} \underbrace{\langle u_0|\epsilon_\beta\rangle\langle\epsilon_\beta|F|E_a\rangle\langle E_a|v_0\rangle}_{\rho_{\beta a}} e^{-i(E_a-\epsilon_\beta)t}. \tag{4.17}$$

All complex residues $\rho_{\beta a}$ and cross terms jointly determine the reading, with $C(0) \neq 1$ in general. This measures frequency differences between the two complete H/B histories; it cannot automatically be replaced by a positive-weight spectral measure of H. Position-resolved readout must also retain position projectors and their cross terms within the same complete evolution. The common field evolution cannot simply be cancelled from the physical-momentum readout.

This distinction can be checked directly from the generators. Since B does not act on nucleon orientation, $[B, Y_A] = [B, Z_A] = 0$. The operator D_A also commutes with both orientations, and $\{Y_A, Z_A\} = 0$. In a fixed reference branch, M is scalar, hence $K^2 = M^2 I + g_{\mathrm{tr}}^2 D_A^2$. Define the two-sided operator $\mathcal{L}(F) = FH - BF$ acting on the switch-off bridge. Expanding it twice gives

$$\begin{aligned} \mathcal{L}^2(F) - M^2 F - g_{\mathrm{tr}}^2 F D_A^2 &= F[K,B] + 2[F,B]K + [[F,B],B], \\ [K,B] &= i g_{\mathrm{tr}} Z_A F_A, \qquad F_A = i[B, D_A]. \end{aligned} \tag{4.18}$$

The original relation between the half-gap and squared transport therefore acquires the field force kernel and finite-time switch-off terms in the actual readout. The same F_A enters the change in physical momentum: with position scale ℓ and $\Pi_A = D_A/\ell$, the plateau relation is $\dot{\Pi}_A = F_A/\ell$. If the numerical interaction is normalized by the energy scale E_*, recovering the physical force requires a further factor of E_*. The last two terms must be retained unless their vanishing has been established; in particular, both vanish when $[F, B] = 0$. Nor can the rest-energy constant of the compensating electron be arbitrarily subtracted from this complete history.

The influence of the same mass arm on motion can be derived directly from the two orientation equations, without invoking a single dispersion curve. Let $\phi(t)$ be the original edge-phase control, and define $C_A = (S_A^\dagger + S_A)/2$ and $D_{A,\phi} = \cos\phi D_A - \sin\phi C_A$. In the two-orientation basis of Y_A, choose phases so that the off-diagonal elements of Z_A equal 1, and denote the complete state components by $\psi_\pm$. Eliminating the negative-orientation component from the actual beginning of the history gives

$$\begin{aligned} i\dot{\psi}_+(t) &= [B(t) + M(t)]\psi_+(t) + g_{\mathrm{tr}}(t) D_{A,\phi(t)} U_-(t,0)\psi_-(0) \\ &\quad - i\int_0^t \Sigma(t,s)\psi_+(s)\, ds, \\ U_-(t,s) &= U_B(t,s)\exp\left(i\int_s^t M(v)\, dv\right), \\ \Sigma(t,s) &= g_{\mathrm{tr}}(t) g_{\mathrm{tr}}(s) D_{A,\phi(t)} U_-(t,s) D_{A,\phi(s)}. \end{aligned} \tag{4.19}$$

The operator $U_B(t,s)$ propagates the actual time-dependent $B(t)$, while $M(t)$ is scalar within a fixed reference branch. The loading interval is included in this time-ordered history. The second term comes from the negative-orientation prehistory. It and the memory kernel follow jointly from block elimination in the complete self-adjoint interaction, sharing the structure of generalized equations of motion that retain the initial source in projection methods. [9]

Differentiate with respect to the control on the original transport edge, and set $C_{A,\phi} = \cos\phi C_A + \sin\phi D_A$. The current vertex at the left endpoint of the memory kernel is

$$-\partial_{\phi_L}\Sigma(t,s) = g_{\mathrm{tr}}(t)g_{\mathrm{tr}}(s)C_{A,\phi(t)}U_-(t,s)D_{A,\phi(s)}. \tag{4.20}$$

Both M from the phase readout and the field propagation B enter the memory kernel for transport. The full control response also includes changes in the right endpoint, state, and prehistory source, and must satisfy continuity for position probabilities, directed-edge currents, and currents across the cut.

§4.4 Constraints on Independent Motion from Finite Readouts

A finite realization retaining all source branches and reference outputs uses a three-site ring, a 1344-dimensional Gauss-constrained matter sector for the proton with a mobile negatively charged electron, and a 288-dimensional neutral-vacuum sector. The source, work, and previous records remain included through the isometry above; these dimensions do not count all supply resources. The parent mass port has finite cutoff $L = 4$ and strength $g_B = .7$. On the plateau, take $\delta = .08$, $\alpha = \pi/4$, and $G_N = 2$, retain the maintenance strength of the real-part reference $\lambda = .0261682266650313$, and include both $X_R = \pm 1$ input branches. The positive-Y_A nucleon preparation and, for the proton process, the positive-Z_C electron preparation are supplied from the actual start of the history. Times below are measured in E_*^{-1}, and generator coefficients are normalized by E_*.

Set $\beta_{\mathrm{arm}} = b_{\mathrm{arm}}/\mu_0 = .015854864758964415$ and $\kappa_{\mathrm{arm}} = a_{\mathrm{arm}}/(\mu_0 + b_{\mathrm{arm}}) = .04553733798100358$, and let $m = \delta^2(\mu_0 + b_{\mathrm{arm}})$ be the arm center of the two neutral reference branches. Then $M_{n,r} = m(1 + r\kappa_{\mathrm{arm}})$ and $M_{p,r} = m[(1 - \beta_{\mathrm{arm}})/(1 + \beta_{\mathrm{arm}}) + r\kappa_{\mathrm{arm}}]$. These two ratios and the independent λ remain conditions of the connection. The field parameters ξ, ζ multiply the auxiliary exchange and compensating-electron transport, respectively, with the actual schedule $B_{N,r}(u) = R_{\mathrm{aux}} + \xi u V_N + u^2(E_{C,\mathrm{mass}} + \zeta E_{C,\mathrm{hop}} + r\lambda I)$; both electron terms vanish in the neutral vacuum.

The numerical deployment uses the directed edges $(0,1), (1,2), (2,0)$, with flux $n_e \in \{-1, 0, 1\}$ on each edge. The charged basis satisfies the Gauss constraint $(\operatorname{div} n)(x) = \mathbf{1}_{x=x_A} - \mathbf{1}_{x=x_C}$, and the neutral basis satisfies $\operatorname{div} n = 0$, with divergence defined as outgoing minus incoming flux. Shift matrix elements leaving the cutoff vanish. The charged position–flux space contains 21 states and the neutral space 9; four two-state auxiliaries, together with the nucleon and electron orientations, give the dimensions above. The auxiliary interaction is $R_{\mathrm{aux}} = 2\sum_{e=1}^{3} Z_e + 2.5Z_f - 8.5I$, and the exchange is $V_N = .8\sum_{e=1}^{3} X_e n_e + 1.1[X_f(I - \operatorname{Re} W) - Y_f \operatorname{Im} W]$, where W raises all three edge fluxes by one and f labels the loop auxiliary.

Initially, all four auxiliaries are in their positive-Z states, the nucleon and electron are at site 0, and all fluxes vanish; the neutral process has no electron. The electron activity terms are $E_{C,\mathrm{mass}} = M_C Y_C$ and $E_{C,\mathrm{hop}} = 2Z_C D_C$, with fixed coefficient $M_C = .00023604226058942325$ inherited from the conditional electron value in the same-source finite model; the actual terms still carry the factor u^2. Here Y_C is the electron mass-orientation operator, anticommuting with the transport orientation Z_C. Loading uses $u(t) = .08(10s^3 - 15s^4 + 6s^5)$, with $s = t/80$, followed by a hold of

78.125. The branches and finite-time switch-offs for each readout are implemented separately as specified below.

The neutral coherence readout allows the mass-arm and transport coefficients to be inferred. Here $S_A^3 = I$, and the three eigenvalues of D_A are $0, \pm\sqrt{3}/2$. The complete B_n commutes with nucleon orientation, position, and transport. The common field propagation therefore cancels from the total coherence in Eq. (4.16) from the actual beginning of the history, while the field itself continues to evolve. The initial nucleon position preparation is $|x_A = 0\rangle$, so each of the three transport spectral weights is $1/3$. The generators used during the original loading and holding commute with D_A, so the evolution preserves these weights. Their nucleon part is $u(t)^2 K_{n,r}/\delta^2$ throughout the prefix, and the complete $B_n(u)$ at each time commutes with $K_{n,r}$. Hence the propagator for this prefix commutes with $e^{-iK_{n,r}s}$, allowing the expectation of the total coherence to be evaluated at the actual initial nucleon orientation. Each reference branch has half-weight, and the initial nucleon orientation is Y_A+. Taking holding time $T = 78.125$, actual switch-off time $\tau = .01$, and $s = T + \tau = 78.135$ gives

$$\begin{aligned} \mathcal{C}_n(m, g_{\mathrm{tr}}) = \frac{1}{6} \sum_{r=\pm 1} \Big[& e^{-iM_{n,r}s} + 2\cos(E_r s) \\ & - 2i \frac{M_{n,r}}{E_r} \sin(E_r s) \Big], \\ E_r = \sqrt{M_{n,r}^2 + \tfrac{3}{4} g_{\mathrm{tr}}^2}. & \end{aligned} \tag{4.21}$$

This expression retains the complete combination of six frequencies without requiring them to be resolved individually. Two orthogonal reference settings give $\mathfrak{f} = (\mathrm{Re}\,\mathcal{C}_n, \mathrm{Im}\,\mathcal{C}_n)$. The nominal center is $\vartheta_0 = (.0014064567706574482, .0128)$, with coherence $.759624083764348 - .100859285874975i$. Within the prior box $|m - m_0|, |g_{\mathrm{tr}} - g_0| \le 10^{-4}$, differentiation of the actual generator with respect to the two parameters gives the central Jacobian and a Hessian bound. Writing $\rho_{\mathrm{box}} = \sqrt{2}\, 10^{-4}$ for the box radius,

$$\begin{aligned} J_{\mathfrak{f}} &= \begin{pmatrix} -7.89691402350 & -34.29418914326 \\ -71.41090769723 & 1.32508838679 \end{pmatrix}, \\ \|D^2 \mathfrak{f}\| &\le s^2(1 + \kappa_{\mathrm{arm}}^2) < 6118, \\ \sigma_{\min}(J_{\mathfrak{f}}) &\ge 34.20, \\ \|\mathfrak{f}(\vartheta) - \mathfrak{f}(\vartheta')\| &\ge (34.20 - 6118\rho_{\mathrm{box}}) \|\vartheta - \vartheta'\| \\ &\ge 33.334784 \|\vartheta - \vartheta'\|. \end{aligned} \tag{4.22}$$

The final inequality follows by integrating along the line segment between the two points and controls the whole box, not only its center. Including errors from the complete history, pointer, phase and timing, statistics, and numerics in each real reading, take $\epsilon = .00015798392711635$. Set $E_{\mathrm{par}} = \sqrt{2}\epsilon/(34.20 - 6118\rho_{\mathrm{box}})$. The two rows of $J_{\mathfrak{f}}^{-1}$ have upper bounds $c = (.014483, .032247)$ in the ℓ^1 norm and $d = (.013955, .029213)$ in the ℓ^2 norm, so componentwise $e_i \le c_i\epsilon + d_i 6118\rho_{\mathrm{box}} E_{\mathrm{par}}$.

For an exact reading $\mathbf{y}$ in the data domain $\|\mathbf{y} - \mathfrak{f}(\vartheta_0)\|_\infty \le \epsilon$, use the fixed-point map $T_{\mathbf{y}}(\vartheta) = \vartheta + J_{\mathfrak{f}}^{-1}[\mathbf{y} - \mathfrak{f}(\vartheta)]$. This map sends the ball of radius 7×10^{-6} around the nominal center into a ball of radius 6.537×10^{-6}, with contraction constant below .001253. Together with injectivity over the whole box, this gives a unique root for

each specified exact reading in the data domain above. The union of these roots over the allowed reading range satisfies the componentwise m, g_{tr} error bounds below. Readings outside the box or incompatible with it require separate examination and cannot be forced into a fit.

Field parameters are constrained by continuity readings of the same actual carrier. Choose $w(t) = 30(t/T)^2(1-t/T)^2/T$, so that $\int_0^T w = 1$ and $\int_0^T |w'| = .048$. Let W denote the original flux loop and define $B_{\mathrm{aux}} = I - \operatorname{Re} W$ and $C_{\mathrm{aux}} = -\operatorname{Im} W$. The corresponding auxiliary interaction is $2.5Z_{\mathrm{aux}} + 1.1\xi u(X_{\mathrm{aux}}B_{\mathrm{aux}} + Y_{\mathrm{aux}}C_{\mathrm{aux}})$. On the plateau $u = \delta = .08$, multiply its Y continuity equation by w and integrate. The endpoint weights vanish, yielding

$$\begin{aligned} \frac{d}{dt}\langle Y_{\mathrm{aux}}\rangle &= 5\langle X_{\mathrm{aux}}\rangle - .176\xi\langle Z_{\mathrm{aux}}B_{\mathrm{aux}}\rangle, \\ F_\xi(\xi) &= .176\xi\left[z_\xi - \frac{q_\xi}{\sin(\pi\xi/2)}\right] - (5x_\xi + .048y_\xi). \end{aligned} \tag{4.23}$$

Here x_ξ, z_ξ are the auxiliary X, Z readings weighted by w, respectively; y_ξ is obtained by sampling with density $|w'|/.048$ and multiplying each Y outcome by $\operatorname{sign}(w')$. The joint port of the loop and Z reads q_ξ, with ideal value $\sin(\pi\xi/2)\int w\langle Z_{\mathrm{aux}} \operatorname{Re} W\rangle$. Its gain varies with the same unknown ξ and has not been replaced by a known constant.

The actual electron position $x_C = 0, 1, 2$ supplies another continuity equation. Let S_C be the electron transport, with flux displacement opposite to that on the positively charged nucleon edge, and set $D_C = (S_C^\dagger - S_C)/(2i)$. The operator Z_C is the electron transport orientation. Take $O_C = Z_C i[D_C, x_C]$ and retain the directed-edge weights across the cut. With plateau electron transport coefficient $g_C\zeta$ and $g_C = .0128$,

$$\begin{aligned} \frac{d}{dt}\langle x_C\rangle &= g_C\zeta\langle O_C\rangle, \\ N_\zeta &= -\int_0^T w'\langle x_C - 1\rangle\, dt, \qquad Q_\zeta = \sin(\pi\zeta/2)\int_0^T w\langle O_C\rangle\, dt, \\ F_\zeta(\zeta) &= g_C Q_\zeta \frac{\zeta}{\sin(\pi\zeta/2)} - N_\zeta. \end{aligned} \tag{4.24}$$

The three outcomes of the implemented electron port, the two Z_C branches, and the final position give 18 outputs. Multiplying the port score $(1, -1, 0)$ by the Z_C sign and final-position weight $(-2, 1, 1)$ gives Q_ζ including the cut contribution; dark records are retained. Both types of instrument are supplied from their respective complete starting states, and all u-dependent interaction terms are switched off at the end, with $\tau_c = .001$. Routing with gain controlled by the same parameter and compensation of the auxiliary phase are explicit conditions. If the gain is independent, it must be retained as an additional parameter.

The table below gives model reading centers after finite-time switch-off. They allow the inversion to be checked; actual inference must use the corresponding measured means.

The total error of each normalized score is $e = .0001623923702081354$, including loading prehistory, the implemented readout instrument, finite-time sampling, and statistical and numerical budgets. The continuity residual caused by switch-off is

Reading	Model center	Center-box half-width
x_ξ	.03441701160826783	e
y_ξ	$-.00015831024552066204$	e
z_ξ	.9990273915777143	e
q_ξ	.02134746337207389	e
N_ζ	.005073159278211831	$.048e$
Q_ζ	.39634051499999534	$2e$

separately controlled by the actual commutator integral. For the prior $\xi \in [.9, 1.1]$, we have $F'_\xi \geq .170918364317$, total residual $E_\xi = .00137683645244$, and nominal root 1.00003365917645. Adding the full reading-center-box drift .005166366454 to the root error radius .008055520880 gives the enclosure below. For $\zeta \in [.5, 1.5]$, the function $\zeta/\sin(\pi\zeta/2)$ is strictly increasing, with $F'_\zeta \geq .00153840555319$. The combined switch-off, data, and numerical residual is $E_\zeta = 1.97034159449 \times 10^{-5}$, with a further $E_{\mathrm{data}} = 1.66136814772 \times 10^{-5}$ for the complete center box. Directly solving $F_\zeta = \pm(E_\zeta + E_{\mathrm{data}})$, rather than replacing it by a loose radius based on the minimum slope over the entire prior, yields

$$\begin{aligned} m &= .0014064567706574482 \ \pm 2.38 \times 10^{-6}, \\ g_{\mathrm{tr}} &= .0128 \ \pm 5.28 \times 10^{-6}, \\ \xi &\in [.986811771843,\ 1.013255546510], \\ \zeta &\in [.992777566682,\ 1.007096241901], \\ \Delta M_{\mathrm{arm}} &= \frac{2\beta_{\mathrm{arm}} m}{1+\beta_{\mathrm{arm}}} E_* = (4.39022987666544 \times 10^{-5} \ \pm 7.43 \times 10^{-8}) E_*. \end{aligned} \tag{4.25}$$

Equation (4.25) gives conditional intervals for the reading-center box and instrument budgets above. Both continuity equations retain the complete histories and gains associated with the unknown parameters; the neutral cancellation is restricted to that total coherence. The physical calibrations of E_* and ℓ remain to be specified separately.

The independent motion uses the nominal parameters frozen in advance and the edge control $\phi(t) = .02\sin(4\pi t/78.125)$, which was not used in parameter estimation. Let $U_{\phi,r}, U_{0,r}$ denote propagation of the same loaded carrier with and without the control. The loading isometry L_r maps the two-dimensional nucleon-orientation input into the complete carrier and includes the prefix from the actual starting point to the onset of motion, with a further platform propagation of duration .01 after loading and holding. The position and field inputs retain their original fixed preparations, and the electron input is prepared in the original Z_C+ state. The operator G_g propagates the finite-time switch-off that brings all u-dependent interaction terms to zero over duration .01 at the end of the motion. The unit-gain endpoint target defined by this actual switch-off and its input difference operator are

$$\begin{aligned} O_D &= G_g^\dagger D_A G_g, \\ \mathcal{T}_D &= \frac{1}{4} \sum_{r=\pm 1} L_r^\dagger \left(U_{\phi,r}^\dagger O_D U_{\phi,r} - U_{0,r}^\dagger O_D U_{0,r} \right) L_r. \end{aligned} \tag{4.26}$$

The factor $1/4$ retains the half-weight of the signal reference arm and the half-weight of each X_R branch; the other reference arm cancels in the difference. The prediction continues to use the complete L_r, including its dependence on $m, g_{\mathrm{tr}}, \xi, \zeta$.

The Gauss-constrained shift S_A is a partial isometry, so its input support is a projector. The three outcome operators of the original momentum port are $K_{\pm} = (I \mp i \sin \vartheta_g S_A)/2$ and $K_d = (I - P_{\mathrm{in}} + \cos \vartheta_g P_{\mathrm{in}})/\sqrt{2}$, where $P_{\mathrm{in}} = S_A^{\dagger} S_A$, with outcome scores $(-1, +1, 0)$, respectively, and $\sum_{j=\pm,d} K_j^{\dagger} K_j = I$. The nominal setting fixes $\vartheta_g = \pi g_{\mathrm{tr}}/(2\widehat{g}_{\mathrm{tr}})$, and $\widehat{g}_{\mathrm{tr}}$ is the frozen transport coefficient. The static readouts then provide three types of bounded joint scores: $\mathcal{C}$ on the coherence axis; $\mathcal{D}$, obtained by multiplying the scores of two momentum ports in the same direction and then the coherence-axis score; and $\mathcal{Q}_{\mathrm{dark}}$, obtained by multiplying the indicator of the first dark record by the coherence-axis score. The latter two are averaged over the two separate settings $S_A, S_A^{\dagger}$, with all outcomes retained; they insert $\sin^2 \vartheta_g D_A^2$ and $[2I - \sin^2 \vartheta_g (S_A^{\dagger} S_A + S_A S_A^{\dagger})]/4$, respectively, into the actual two-history bridge. The complete coherence score is not multiplied by the signal half-weight a second time.

The target and all score operators are compressed to the two-dimensional input space above, retaining arbitrary nucleon orientation and all correlations left by prior interactions; the following matrix relation does not extend to arbitrary inputs in the 1344-dimensional space. Expanding the specified theoretical target in these input operators gives the fixed relation at the preassigned readout times $t_a = 73.71338280344123$ and $t_b = 77.27455695883035$:

$$\begin{aligned} \mathcal{T}_D = {} & 4.397887840043 \times 10^{-5} I \\ & - 1.226288900384 \times 10^{-4} \mathcal{D}_{t_a, X} \\ & + 1.365429761910 \times 10^{-5} \mathcal{Q}_{\mathrm{dark}, t_b, X} \\ & - 1.482921168749 \times 10^{-4} \mathcal{C}_{t_b, Y} + \mathcal{R}_D. \end{aligned} \tag{4.27}$$

The sum of the absolute values of the three weights is $2.845753045324 \times 10^{-4}$. The model centers of the three scores are .32174429129511317, .06778647571092997, and $-.06582428757151292$, respectively; substitution into the relation above reproduces the center of the endpoint target. The weights are determined by the frozen theoretical interaction and the static instruments, without fitting the motion readouts. The remainder $\mathcal{R}_D$ retains parameter variation throughout the complete history: the second-order field polynomial is bounded by its matrix Bernstein coefficients over the full parameter box, while higher orders are controlled by common-source Duhamel insertions and a change of frame using the auxiliary gap. Mass–field cross terms are included. The nominal momentum difference is $\Delta D_{\mathrm{read},0} = 1.521053274245407 \times 10^{-5}$; its errors are propagated through the actual relation above as follows.

The actual momentum does not commute with field propagation, so the endpoint readout must still be connected back to the plateau. The integral identity for the switch-off bridge and the bound over the full parameter/readout box give

$$\begin{aligned} \Gamma_D = G_g^{\dagger} D_A G_g - D_A &= \int_0^{.01} U_g(s)^{\dagger} i [H_g(s), D_A] U_g(s)\, ds, \\ \ell \Delta \Pi_p = \Delta D_{\mathrm{read}}^{\mathrm{ideal}} - \Delta \Gamma_D, &\qquad |\Delta \Gamma_D| \le 7.5 \times 10^{-7}. \end{aligned} \tag{4.28}$$

Gain, statistical, and other deviations of the actual reading from the unit-gain target $\Delta D_{\mathrm{read}}^{\mathrm{ideal}}$ are included in the table above. Here $\Delta \Gamma_D$ is the difference between the complete controlled and zero-control histories, with the same reference weights

Error source	Absolute bound on ΔD_{read}
Full m, g_{tr} contribution and its cross terms with the field	$4.241002134 \times 10^{-6}$
Variation of the static scores with m, g_{tr}	$4.078388399 \times 10^{-7}$
Remainder of the second-order matrix relation over the full box	$6.531045348 \times 10^{-7}$
Third-order field remainder of the motion target	$3.464793981 \times 10^{-6}$
Third-order field remainder of the static scores	$7.647865325 \times 10^{-8}$
Field-parameter double difference at higher control orders	$4.014272212 \times 10^{-7}$
Remaining full-history, statistical, and gain budget of the momentum instrument	$4.013882098 \times 10^{-6}$
Numerical error in the state columns, static gain, center box, and analytic/floating-point allowance	$4.054626261 \times 10^{-7}$
Total bound (summed before rounding the individual terms)	$1.366399008918151 \times 10^{-5}$

retained. The original neutral transport commutes with the complete $H_n(t)$, so its ideal $\Delta\Pi_n = 0$, although its position currents may still change. Adding the total momentum-difference bound and the switch-off bound gives the common-source plateau comparison

$$\ell(\Delta\Pi_p - \Delta\Pi_n) \in [.796542653,\ 29.624522832] \times 10^{-6}. \tag{4.29}$$

This positive impulse difference is a model prediction under the same underlying joint interaction and a compatible readout box. A comparison of the two actual instruments must also include the neutral instrument's budget. The statistical events used here are not assumed independent; their joint coverage has a lower bound of 94%. The ordinary floating-point centers and analytic allowances have not yet been certified by outward rounding. Each setting is implemented separately and accounts for preparation, control, readout, and failure resources throughout its own complete history.

The inversion above constrains the mass arms, the field parameters, and their memory contribution to motion. Identifying the complete rest mass further requires preserving the nucleon's spatial pattern and correlated internal readings within a finite window and tolerance. Position coherence, directed currents, currents across the cut, and a common reference must then determine an identifiable spectral cluster and its phase center, with the same spectral combination entering the motion kernel in Eq. (4.19). The internal field and compensating matter may remain active.

§4.5 Mass Splittings within Baryon Multiplets

Extending the flavor carrier to u, d, s gives the three-factor flavor decomposition $\mathbf{3}^{\otimes 3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$. This flavor classification uses the $SU(3)$ representation structure of the Eightfold Way [10, 11] and acts on a different factor from the

color singlet in Chapter 2. When the internal spatial ground state belongs to the symmetric representation, color antisymmetry requires the joint flavor–spin state to be symmetric. The corresponding decomposition is

$$\mathrm{Sym}^3\big(\mathbb{C}^3_{\mathrm{flavor}} \otimes \mathbb{C}^2_{\mathrm{spin}}\big) \cong (\mathbf{10}, \mathbf{4}) \oplus (\mathbf{8}, \mathbf{2}). \tag{4.30}$$

This decomposition agrees with the organization of the standard spin–flavor $SU(6)$ representation. [12] Here **4** and **2** are the spin-3/2 and spin-1/2 representations, respectively. The mixed flavor symmetry of the octet must be combined with mixed spin symmetry to form a jointly symmetric state, as in the nucleon combination in Eq. (2.8). Thus, the two octets in the flavor tensor product do not represent two independent families of spatial-ground-state nucleons. Strange-flavor number and internal spin alter participation in the response, giving different rest frequencies to members supported by the same three-color singlet.

Standard flavor-symmetry analysis obtains mass relations within multiplets through symmetry breaking. [13] In the common mass function used here, different members have different effective participation weights, with the leading-variance parametrization of §4.1 used to specify the spectral preparation. Let n_s be the strange-flavor number and j the total baryon spin, and write the effective spectral parameter as ξ_{eff}:

$$\begin{aligned} \xi_{\mathrm{eff}} &= 3 + n_s + \frac{n_s(n_s-1)}{4} + 2\,\mathbf{1}_{j=3/2}, \\ \beta_\xi &= 1 + \xi_{\mathrm{eff}}^{-1/2}, \\ M^{(0)}_{\mathcal{B}} &= E_0 \sqrt{\frac{Z_L(\beta_\xi)}{C_{\mathrm{bg}}}}\, \log Z_L(\beta_\xi). \end{aligned} \tag{4.31}$$

This spectral preparation assigns participation weights to a single strange flavor, a strange-flavor pair, and the spin-3/2 branch, allowing different members to be compared through the same mass function in Eq. (4.3). These weights are summarized by the effective parameter ξ_{eff}; the color factor of every member remains the three-factor singlet ε_3.

Multiplet member	n_s	j	ξ_{eff}
N	0	1/2	3
Λ, Σ	1	1/2	4
Ξ	2	1/2	11/2
Δ	0	3/2	5
Σ^*	1	3/2	6
Ξ^*	2	3/2	15/2
Ω	3	3/2	19/2

Table 4-1. Participation parameters in the selected extension of the spectral preparation. These values enter the common mass function; the remaining splittings between members are read separately from the internal interaction.

At fixed $L \geq 2$ and C_{bg}, increasing ξ_{eff} lowers β_ξ and therefore increases Z_L; the function $\sqrt{Z}\log Z$ is strictly increasing on $Z > 1$. Equation (4.31) thus gives

a hierarchy of collective masses that rises with strange-flavor participation and the spin branch. The states Λ and Σ share the same spectral baseline, while their difference lies in the internal pair interaction.

The contribution of spin-dependent interactions to the $\Sigma-\Lambda$ splitting was treated in early work on the gauge quark model. [14] Consider Λ and Σ^0, both with flavor composition uds. If the relevant internal response is $H_{ud} = \kappa_{ud}\boldsymbol{\sigma}_u \cdot \boldsymbol{\sigma}_d$, its matrix elements for pair spins 0 and 1 are $-3\kappa_{ud}$ and $+\kappa_{ud}$, respectively. At an order where the other contributions are equal, extending the result for Σ^0 to the Σ multiplet in the isospin-symmetric limit gives

$$\Delta M_{\Sigma\Lambda} = 4\kappa_{ud}, \qquad M_{\mathcal{B}} = M_{\mathcal{B}}^{(0)} + \delta M_{\mathcal{B}}^{\mathrm{int}}. \tag{4.32}$$

Equation (4.32) locates the splitting coefficient in an actual pair-induced frequency shift, whose value is calculated from the selected internal exchange. Spectral preparation and internal pairing thus control the common multiplet hierarchy and the splittings within a level, respectively; their contributions must be added using the same complete timing convention.

§4.6 Baryon Excitations

Excited states are identified along the same carrier after formation. When a nucleon acquires a nonzero propagation wave number, Eq. (3.7) changes its motional energy; if its mode, internal state, and return frequency still belong to the original rest branch, it remains a moving nucleon of the same kind. A new baryon excitation instead requires a different rest response of the complete internal interaction, together with an actual matrix element connecting the original preparation to that response.

The single-port interaction in Eq. (3.5) has only $+r_B$, $-r_B$, and a zero dark spectrum. The positive and negative bright branches have eigenvalues of equal magnitude; mass and transport orientation are carried by Y_A, Z_A on a separate factor. Changing the sign of the bright state alone does not produce a new positive mass scale, nor does the zero dark spectrum automatically identify a baryon excitation. Further excitations must be obtained from the extended actual internal and mode interactions. For an established self-adjoint rest generator H_{rest}, the coherent return of a prepared state Ψ has the spectral representation

$$C_\Psi(t) = \langle \Psi, e^{-itH_{\mathrm{rest}}}\Psi\rangle = \int e^{-itE}\, d\mu_\Psi(E), \qquad \mu_\Psi(\Delta) = \langle \Psi, P_{H_{\mathrm{rest}}}(\Delta)\Psi\rangle. \tag{4.33}$$

Here $P_{H_{\mathrm{rest}}}$ is the spectral projection-valued measure of the rest generator; the resulting scalar spectral measure records both the accessible energies and their overlaps. This representation presupposes identification of the actual carrier at rest and of this generator; the finite-time switch-off H/B instrument of §4.3 cannot be replaced by such a single-generator measurement merely by writing a spectral expansion. Discrete rest components give definite frequencies. Components connected to actual exit channels must also be treated together with the survival readouts of Chapter 3. The preceding paper, *The Mechanism of Decay from the Bost–Connes System*, establishes the relation among return phase, departing share, and total elapsed duration. [15] Within a window of approximately exponential survival, the

same return signal can identify a resonance center and width, connecting with the standard approximate descriptions of natural linewidths and resonance line shapes. [16, 17] General oscillations in a finite system retain their full time dependence.

This description distinguishes mass comparisons between spectral preparations from excitations under a single interaction. Equation (4.31) organizes the preparations of different members, whereas Eq. (4.33) determines the components that the corresponding interaction can excite and read out within a given complete system. Locating individual excitation peaks further requires incorporating the relevant internal exchanges into the same rest interaction. Electromagnetic and weak probes select the visible components through the response matrix elements considered next.

§5 Electroweak Response of Nucleons

Electroweak probes act on the internal structure of the same nucleon. A magnetic field changes the phases of spin branches; the charged weak interaction connects protons and neutrons and transfers energy and charge to the leptonic output. The nucleon states in Chapter 2 supply the internal matrix elements, while the actual contact interaction and its return determine the normalization of measurable responses. The weak-response model of the preceding work thus connects nucleon structure to decay outputs. [3, 15]

§5.1 Nucleon Magnetic Moments

In a weak uniform magnetic field B_{ext}, the magnetic moment is defined by the linear frequency shift of the nucleon spin branches. Compare the same orbital and spatial preparation, using the same time and energy calibration as for the zero-field mass. With common center $\bar{E}_N(B_{\mathrm{ext}}) = \frac{1}{2}(E_{N,\uparrow} + E_{N,\downarrow})$ and $\bar{E}_N(0) = m_N$, write

$$\begin{aligned} E_{N,\uparrow}(B_{\mathrm{ext}}) &= \bar{E}_N(B_{\mathrm{ext}}) - \mu_N B_{\mathrm{ext}} + O(B_{\mathrm{ext}}^2), \\ E_{N,\downarrow}(B_{\mathrm{ext}}) &= \bar{E}_N(B_{\mathrm{ext}}) + \mu_N B_{\mathrm{ext}} + O(B_{\mathrm{ext}}^2). \end{aligned} \tag{5.1}$$

The magnetic moment is therefore read from the field-induced phase difference between the spin branches; a common orbital shift cancels in their difference. Orbital responses, exchanges involving multiple internal slots, and the environment left by prior interactions are included in the complete frequency shift whenever they contribute to the phase. To separate the contribution of internal structure, first take the leading single-slot magnetic response. Its internal operator and flavor-component matrix elements are

$$\widehat{\mu}_z^{(1)} = \sum_{a=1}^{3} \big(\mu_u P_u^{(a)} + \mu_d P_d^{(a)}\big)\sigma_z^{(a)}, \qquad \Delta f_N = \langle \Phi_{N,\uparrow}| \sum_a P_f^{(a)} \sigma_z^{(a)} |\Phi_{N,\uparrow}\rangle. \tag{5.2}$$

Here $P_f^{(a)}$ projects onto flavor f at internal slot a, and μ_u, μ_d are the leading frequency-shift coefficients of the two flavor branches under the same external magnetic interaction. To retain flavor–spin correlations, write $\Psi_a = \phi_{p,a} \otimes \eta_a$ $(a = \rho, \lambda)$ and set $O_f = \sum_j P_f^{(j)} \sigma_z^{(j)}$. Substitution of the permutation vectors in Chapter 2 gives the compressed matrix elements on $(\Psi_\rho, \Psi_\lambda)$:

$$\left[\langle\Psi_a|O_u|\Psi_b\rangle\right]_{a,b} = \begin{pmatrix} 1 & 1/3 \\ 1/3 & 1 \end{pmatrix}, \qquad \left[\langle\Psi_a|O_d|\Psi_b\rangle\right]_{a,b} = \begin{pmatrix} 0 & -1/3 \\ -1/3 & 0 \end{pmatrix}. \tag{5.3}$$

The nucleon state is an equal-weight coherent sum of the two components, so the cross terms also contribute to the expectation values, giving $\Delta u_p = 4/3$ and $\Delta d_p = -1/3$; the neutron interchanges the two. Hence

$$\boxed{\mu_p^{(1)} = \frac{4\mu_u - \mu_d}{3}, \qquad \mu_n^{(1)} = \frac{4\mu_d - \mu_u}{3}.} \tag{5.4}$$

The joint flavor–spin permutation structure determines the coefficients $4/3$ and $-1/3$. When the two flavors share a single-slot magnetic response normalized in proportion to charge, $\mu_u = 2\mu_{\text{mag}}/3$ and $\mu_d = -\mu_{\text{mag}}/3$, giving $\mu_p^{(1)} = \mu_{\text{mag}}$ and $\mu_n^{(1)} = -2\mu_{\text{mag}}/3$, with ratio $-3/2$. This agrees with the standard $SU(6)$ analysis of electromagnetic responses. [18, 19] Although the neutron has zero total charge, correlations between its internal charged components and spin give a nonzero magnetic moment.

The complete magnetic response also includes the participating orbital and multiple-slot interactions and their returns. Components outside the four nucleon states generated by the single-slot operator must likewise be retained in the return calculation. The electromagnetic contact determines the absolute normalization and corrections; if the two nucleon responses share the same multiplicative return factor, that factor cancels in their magnetic-moment ratio.

§5.2 Nucleon Axial Response

The axial response connects nucleon spin to weak transitions. On the flavor doublet, take $\tau_3 u = u$ and $\tau_3 d = -d$. For the internal states of Chapter 2, the proton matrix element of the sum of single-slot spin–flavor operators is $\langle\sum_a \tau_3^{(a)}\sigma_z^{(a)}\rangle_p = 5/3$. This number reflects the internal arrangement of the specified state; the measurable axial coupling also includes the weak-interaction endpoints, returns, and normalization relative to the vector response. The relation between the axial coupling and strong-interaction scattering information is expressed by the Adler and Weisberger sum rules. [20, 21]

Here we use the low-energy return model of Ref. [15], already normalized to the vector response $g_V = 1$, and denote the complete axial-to-vector response ratio by the positive number g_A. The value $5/3$ above retains its meaning as an internal single-slot matrix element and is not multiplied into this model again. That work uses in-phase returns from color insertions, with a factor $a = 3\alpha_s^{(0)}/2$ per insertion and a single color-singlet endpoint term $\alpha_s^{(0)}/\sqrt{3}$. For $|a| < 1$, the total return satisfies

$$R_A = 1 + aR_A, \qquad \boxed{g_A = R_A + \frac{\alpha_s^{(0)}}{\sqrt{3}} = \frac{1}{1 - 3\alpha_s^{(0)}/2} + \frac{\alpha_s^{(0)}}{\sqrt{3}}.} \tag{5.5}$$

Equation (5.5) adds the in-phase amplitude of repeated returns to the single endpoint response. Using the leading color response $(\alpha_s^{(0)})^{-1} = 8.701707448621\ldots$ of

the preceding work gives $g_A = 1.274633018\ldots$. This ratio enters the weak-transition amplitude directly.

Flavor conversion by the charged current uses the normalized two-generation link of the preceding work; its normalized two-component structure corresponds to Cabibbo mixing. [22] Let the normalized amplitude of the formation link be λ_C. Then

$$|V_{us}| = \lambda_C = 0.2247147851\ldots, \qquad |V_{ud}| = \sqrt{1-\lambda_C^2} = 0.974424582\ldots. \tag{5.6}$$

Equation (5.6) incorporates the formation link into the selected two-generation charged current. With this normalization, Eqs. (5.5) and (5.6), together with $g_V = 1$, determine the nucleon weak response used below.

§5.3 Neutron Weak Decay

Free-neutron weak decay is $n \to p + e^- + \bar{\nu}_e$. The internal flavor change transfers charge and energy to the leptonic output, while the modes and internal states of the initial and final nucleons are those of Chapter 2. Reference [15] provides the allowed-transition calculation at small momentum transfer; we use its low-energy vertex and complete external-state normalization directly:

$$\begin{aligned} \mathcal{M}_\beta &= \frac{G_F V_{ud}}{\sqrt{2}} \left[\bar{u}_p(\gamma^a - g_A\gamma^a\gamma^5)u_n\right]\left[\bar{u}_e\gamma_a(1-\gamma^5)v_\nu\right], \\ G_F &= \frac{1}{\sqrt{2}v_{\mathrm{EW}}^2} = 1.16664665\ldots\times 10^{-5}\ \mathrm{GeV}^{-2}. \end{aligned} \tag{5.7}$$

Here u_p, u_n, u_e, v_ν are the corresponding external-state spinors and γ^a are the matrices in this low-energy representation. This representation takes the effective V−A form, whose chiral structure is described in the foundational weak-interaction literature; [23, 24] the response coefficients and time normalization follow Ref. [15]. With the continuum three-body external-state normalization used there, the total width is $\Gamma_n = (2m_n)^{-1}\int d\Phi_3\,\overline{|\mathcal{M}_\beta|^2}$, where the overline denotes averaging over the initial spin and summing over final spins.

After removing the common external-state factors, the leading nucleon currents are $J_V^0 = \chi_p^\dagger\chi_n$ and $J_A^i = g_A\chi_p^\dagger\sigma_i\chi_n$, where χ_N are the two-component nucleon spin states. The spin sum and angular integration over unobserved directions then give $1 + 3g_A^2$; the factor 3 comes from the three spatial components of the axial vector.

Neglecting recoil, electromagnetic corrections, and the neutrino mass in the total width, let $E_\beta = \Delta m$ denote the endpoint of the electron total energy and write $p_e = \sqrt{E_e^2 - m_e^2}$. Then

$$\frac{d\Gamma_n}{dE_e} = \frac{G_F^2|V_{ud}|^2}{2\pi^3}(1+3g_A^2)\,p_e E_e (E_\beta - E_e)^2, \qquad m_e \le E_e \le E_\beta. \tag{5.8}$$

The electron and antineutrino share the available energy, producing a continuous spectrum. [25] The total-energy endpoint E_β and electron kinetic-energy endpoint $Q_\beta = \Delta m - m_e$ correspond to different readings; finite recoil modifies the exact endpoints at higher order. With $w = E_e/m_e$ and $w_0 = \Delta m/m_e$, the total width is

$$\boxed{\begin{aligned} f_0(w_0) &= \int_1^{w_0} w\sqrt{w^2-1}(w_0-w)^2\,dw, \\ \Gamma_n &= \frac{G_F^2|V_{ud}|^2 m_e^5}{2\pi^3}(1+3g_A^2)f_0(w_0). \end{aligned}} \tag{5.9}$$

Using the mass difference in Eq. (4.13) and the electron mass $m_e = 0.5110122512\ldots$ MeV from the same set of inputs in the preceding work gives $f_0 = 1.72268806\ldots$ and $Q_\beta = 0.79261825\ldots$ MeV. Restoring seconds, the mean lifetime is $\overline{T}_n = \hbar/\Gamma_n = 895.71$ s. The lifetime and energy spectrum follow from the same masses and weak responses, inheriting the phase–mass prescription and self-energy normalization specified in Chapter 4 and the effective weak-response conditions of this section. Recoil and electromagnetic corrections can be included within the same decay process as in Ref. [15]; the standard analysis of radiative corrections to allowed β spectra and lifetimes is given in Ref. [26].

The spin-dependent angular distribution also contains interference between the vector and axial responses. Define A_n as the coefficient of $\boldsymbol{P}_n \cdot \boldsymbol{p}_e/E_e$ in the angular distribution, where $\boldsymbol{P}_n$ is the neutron polarization vector. With the same positive-g_A convention, the leading correlation formula for allowed β decay gives [27]

$$A_n = -\frac{2g_A(g_A-1)}{1+3g_A^2} = -0.119187\ldots. \tag{5.10}$$

The mass difference sets the energy window for weak decay; the axial and vector responses set its amplitude. Integration over the complete final state gives the departure rate, while the angular distribution further tests interference between the two responses. The common internal structure of the nucleon remains consistent across these readings, and the remaining outputs are retained as actual decay products.

§6 Interactions between Nucleons

The interaction between two nucleons involves joint evolution of the emitting endpoint, the propagating mediator, and the absorbing endpoint. During exchange, the nucleons change internal flavor or spin, while the propagating component carries the corresponding energy and charge. Absorption returns the joint state to the two-nucleon subspace with no π. *The Mechanism of Fields from the Bost–Connes System* implements this process as a complete nucleon–meson interaction. [7] We use that construction to connect the internal structure to the channels and spatial response of the nuclear force.

§6.1 Formation of Meson Exchange

The π meson uses a color singlet formed from the fundamental color representation and its conjugate, a flavor triplet, and a spin singlet. In the representation of Ref. [3], its color and flavor vectors are

$$\Omega_c = \frac{1}{\sqrt{3}} \sum_{a=1}^{3} |a\bar{a}\rangle,$$
$$\phi_{\pi^+} = |u\bar{d}\rangle, \qquad \phi_{\pi^-} = |d\bar{u}\rangle, \qquad \phi_{\pi^0} = \frac{|u\bar{u}\rangle - |d\bar{d}\rangle}{\sqrt{2}}. \tag{6.1}$$

With zero relative orbital angular momentum and a negative product of the intrinsic parities of the constituent and anticonstituent, this mode has $J^P = 0^-$. Its internal representation permits nucleon flavor exchange while preserving the color singlet. The mass interaction in Ref. [3] and its Supplement S4.4 uses a product preparation of the collective and flavor factors. Write the dimensionless readings as $b_B = M_B/E_0$, $b_u = m_u/E_0$, and $b_d = m_d/E_0$, retaining the product output column and its norm normalization specified there. The two orthogonal flavor outputs are added first; three equal-strength reversible rearrangement channels then give the total squared response $S_\pi = 3b_B(b_u + b_d)$. The bright-state frequency of the corresponding complete response column is proportional to $\sqrt{S_\pi}$. Calibrating the full-cycle drive of the mass interaction to E_0 gives $m_\pi = E_0\sqrt{S_\pi}$, or

$$m_\pi^2 = 3M_B(m_u + m_d), \qquad m_\pi = 138.289476310587\ldots \text{ MeV}. \tag{6.2}$$

Equation (6.2) uses the same input set $m_u + m_d = 6.805667343664\ldots$ MeV and gives a common π mass before electromagnetic splitting. The three rearrangement channels are orthogonal outputs of the complete interaction. Their response columns and coherent returns follow the preceding work and are counted separately from the three coherent components of the color singlet. [3]

A nucleon endpoint must preserve the entire flavor–spin carrier. Let J_N be the isometric embedding of the four nucleon states $(p\uparrow, p\downarrow, n\uparrow, n\downarrow)$, with the common mode and color factors suppressed. In the permutation convention of Chapter 2, take $J_N|p\uparrow\rangle = \Phi_{p,\uparrow}$ and $J_N|n\uparrow\rangle = -\Phi_{n,\uparrow}$, with the spin-down states generated by the same total-spin lowering operator. This phase choice makes the nucleon doublet transform under the standard isospin matrices. Denote the sums of the Pauli operators over the three internal slots by $\mathsf{S}_i = \sum_a \sigma_i^{(a)}$ and $\mathsf{T}_b = \sum_a \tau_b^{(a)}$; the corresponding rotation generators are half these sums. The collective endpoint of the preceding work satisfies the full intertwining relations [7]

$$\mathsf{S}_i J_N = J_N(I \otimes \sigma_i), \qquad \mathsf{T}_b J_N = J_N(\tau_b \otimes I),$$
$$\mathsf{S}_i \mathsf{T}_b J_N = J_N(\tau_b \otimes \sigma_i). \tag{6.3}$$

This interaction maps the complete nucleon back into the same four-state space. It uses the product of total spin and total flavor, including terms involving different internal slots, and therefore has a different scope from the compressed single-slot matrix elements of Chapter 5. To connect this endpoint to charged π outputs, take the Cartesian flavor basis $|\varphi_b\rangle = \sum_{r,s=u,d}(\tau_b)_{rs}|r\bar{s}\rangle/\sqrt{2}$ and write $\tau_\pm = (\tau_1 \pm i\tau_2)/2$. The source column in Supplement D5.1 of Ref. [7] gives

$$\sum_b \tau_b \otimes |\varphi_b\rangle = \sqrt{2}\,\tau_- \otimes |\pi^+\rangle + \sqrt{2}\,\tau_+ \otimes |\pi^-\rangle + \tau_3 \otimes |\pi^0\rangle. \tag{6.4}$$

Here $|\pi^\pm\rangle, |\pi^0\rangle$ denote the flavor vectors of Eq. (6.1); the common color and spin singlets are suppressed. Lowering the nucleon charge is paired with positive-π

emission, and raising it with negative-π emission, so the joint emission preserves total charge.

Next, connect two equal-strength, opposite-phase spatial arms at neighboring positions. Let U_i be a unitary translation by lattice spacing ℓ on a finite three-dimensional periodic lattice, with the orientation convention $U_i|k\rangle = e^{ik_i\ell}|k\rangle$. Set $D_i = (U_i - U_i^\dagger)/(2i\ell)$, and let δ_x be a normalized position-basis state. Here D_i is divided by a length and is distinguished from the dimensionless D_A of Chapter 4. In the absence of flux, the two conventions describe the same direction through $U_i = S_A^\dagger$; charged flux shifts remain part of their respective complete carriers. For a four-state nucleon vector χ, the complete emission column and its momentum-space vertex are

$$\begin{aligned} V_x\chi &= g_\pi \sum_{b,i} (\tau_b \otimes \sigma_i)\chi \otimes |\varphi_b, \chi_\pi^+; 0\rangle \otimes D_i\delta_x, \\ V_b(k) &= g_\pi \tau_b\, \boldsymbol{\sigma}\cdot\widehat{\boldsymbol{k}}, \qquad \widehat{k}_i = \frac{\sin(k_i\ell)}{\ell}. \end{aligned} \tag{6.5}$$

Here χ_π^+ is the positive-eigenvalue state of the mass interaction, with charge distinguished by the flavor vector φ_b; 0 labels the scalar auxiliary component for spatial propagation. The two spatial arms share the internal output and auxiliary record, while their position components remain in an opposite-phase coherent superposition. Fourier transformation produces an odd momentum factor, and the adjoint $V_x^\dagger$ implements absorption. The dimension of g_π is energy times length, so $V_b(k)$ has the dimension of energy. Equal strength along the three axes and the covariant normalization of the three π components follow the preceding construction. Setting $x = x_A, x_B$ in turn and retaining the other nucleon factor gives the emission interactions V_A, V_B at the two endpoints, used below.

§6.2 Channel Structure of the Nuclear Force

In the finite construction of the preceding work, both nucleons and the sectors with zero or one π are included jointly from the outset. Let $\mathcal{H}_{0\pi}$ denote the zero-π subspace and $\mathcal{H}_{1\pi}$ the subspace containing a π and its transport auxiliaries. The emission interactions of the two nucleons are $V_A, V_B : \mathcal{H}_{0\pi} \to \mathcal{H}_{1\pi}$. The common joint generator and its return correction are

$$\boxed{\begin{aligned} H_{\text{joint}} &= \begin{pmatrix} H_{NN} & V^\dagger \\ V & H_{1\pi} \end{pmatrix}, \\ V &= V_A + V_B, \\ \Sigma(E) &= V^\dagger (E - H_{1\pi})^{-1} V. \end{aligned}} \tag{6.6}$$

Here H_{NN} is the two-nucleon interaction in the zero-π space, while $H_{1\pi}$ retains both nucleons, the π mass and position, and the auxiliaries that implement propagation. When E lies outside the spectrum of $H_{1\pi}$, elimination of the π-containing component writes the same joint equation in the return form above; the complete interaction still accounts for energy, charge, and source backreaction. This energy-dependent return form is an instance of projection-based elimination. [28] Expanding Eq. (6.6), the two self-return terms correct the individual nucleon phases, while

the two cross-return terms generate the interaction between nucleons. Paths not distinguished by the complete records add at the amplitude level.

At leading static order, subtract the common two-nucleon rest-energy baseline from both blocks and take the nucleon spin–flavor labels to be degenerate. Actual mass splittings and recoil remain in H_{NN} and $H_{1\pi}$ of Eq. (6.6). The π mass and spatial interaction follow Supplement D5.2 of Ref. [7]. Write $\mathcal{D}_x = (U_1 - I, U_2 - I, U_3 - I)^{\mathsf{T}}$. Spatial propagation is implemented by coupling one scalar auxiliary component to three gradient auxiliary components, with generator

$$\begin{aligned} h_\pi &= \mathcal{M}_\pi \otimes I_4 + \nu \Xi_\pi \otimes \begin{pmatrix} 0 & \mathcal{D}_x^\dagger \\ \mathcal{D}_x & 0 \end{pmatrix}, \\ \mathcal{L}_x &= \mathcal{D}_x^\dagger \mathcal{D}_x = \sum_i (2I - U_i - U_i^\dagger). \end{aligned} \tag{6.7}$$

Here $\mathcal{M}_\pi$ is the original mass exchange, Ξ_π is a self-adjoint unitary orientation operator that anticommutes with it, and ν is the energy scale of position transport. On the mass branch accessible to the source, $\mathcal{M}_\pi^2 = m_\pi^2 I$. Thus h_π^2 restricted to the scalar entrance is $m_\pi^2 + \nu^2 \mathcal{L}_x$. Using $(E - h_\pi)^{-1} = (E + h_\pi)(E^2 - h_\pi^2)^{-1}$ and taking its matrix element in the mass state χ_π^+ at the scalar entrance gives

$$R_\pi(E) = \frac{E + m_\pi}{E^2 - m_\pi^2 - \nu^2 \mathcal{L}_x}, \qquad R_\pi(0) = -m_\pi (m_\pi^2 + \nu^2 \mathcal{L}_x)^{-1}. \tag{6.8}$$

The propagation term $\mathcal{L}_x$ is built from the neighboring-position difference $U_i - I$, whereas the endpoint factor D_i is the centered difference generated by the two opposite-phase arms; they arise from the propagation and emission interactions, respectively. In a static, homogeneous, inversion-symmetric background with equal masses for the three π components, take equal real strengths at the two endpoints and set $G = (m_\pi^2 + \nu^2 \mathcal{L}_x)^{-1}$. Substitution of the complete emission column in Eq. (6.5) into Eq. (6.6) gives the two cross returns [7]

$$\boxed{V_{AB} = -2 g_\pi^2 m_\pi (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) \sum_{ij} \sigma_A^i \sigma_B^j \langle D_i \delta_{x_A}, G D_j \delta_{x_B} \rangle.} \tag{6.9}$$

The factor 2 comes from cross returns in the two directions; all three π exchanges are included in the isospin inner product. Equation (6.9) retains finite positions and the selected periodic boundary conditions. Relative to the accessible mediator gap m_π, the static second-order approximation uses $|E|/m_\pi \ll 1$ and $\|V\|/m_\pi \ll 1$ and requires nucleon motion to be sufficiently slow compared with the mediator response. Here V is the complete emission interaction in the finite joint space, including spatial gradients and internal matrix elements. Its spin and isospin structure determines attraction and repulsion in the different channels.

Comparing endpoint strengths also involves the affine-channel response of the preceding work. [3] For 42 orthogonal complete outputs from the same matched input, each channel has layer weight $1/\mathcal{L}_7$ and directional response $\mathcal{W}_1$, giving

$$\alpha_{\text{nuclear}} = \frac{42}{\mathcal{L}_7} \mathcal{W}_1 = 14.097207871478\ldots, \qquad \mathcal{L}_7 = \frac{3}{2|\zeta(1/2)|}. \tag{6.10}$$

Here $\mathcal{W}_1 = 0.344759924154\ldots$ is the directional response obtained in the preceding work at the fundamental-state spectral point. Equation (6.10) gives the total

response ratio for that preparation. Using it to calibrate the strength of g_π further requires matching it to the actual π endpoint with consistent spatial normalization, units, and total elapsed duration.

§6.3 Spatial Response of the Nuclear Force

Spatial separation enters the nuclear interaction through the actual propagation kernel. The distinction between an outer region governed by meson exchange and an inner region requiring separate treatment was already explicit in early nuclear-force research. [29] In the long-wavelength approximation, consider separations much larger than the lattice spacing, with negligible contributions from periodic images. For $m_\pi/\nu \ll 1$, write $c_\pi = \nu\ell$ and $\kappa = m_\pi/c_\pi$. With continuum point-source normalization, the finite propagation matrix element satisfies $G(x_A, x_B) \simeq \ell^3 Y(r)$, where

$$Y(r) = \frac{e^{-\kappa r}}{4\pi c_\pi^2 r}, \qquad \kappa^{-1} = \frac{c_\pi}{m_\pi}. \tag{6.11}$$

Here $r = |x_A - x_B|$, and c_π is the long-wavelength propagation coefficient of this position interaction; setting $c_\pi = 1$ uses the same calibration as the common speed of light. Equation (6.11) has Yukawa-type exponential decay, expressing the standard connection between mediator mass and finite interaction range. [30] Here the kernel follows from the finite-position propagation of the preceding work, and the two gradient endpoints give $\langle D_i \delta_{x_A}, G D_j \delta_{x_B} \rangle \simeq -\ell^3 \partial_i \partial_j Y(r)$. The finite lattice, boundaries, and response near zero separation continue to be described by Eq. (6.9).

For a propagation kernel depending only on the distance r, the second spatial derivative satisfies $\partial_i \partial_j Y = (Y'' - Y'/r)\hat{r}_i \hat{r}_j + (Y'/r)\delta_{ij}$. This identity separates the derivatives of Eq. (6.11) into radial and tangential parts. [7] With continuum point-source strength $g_c = g_\pi \ell^{3/2}$, one obtains, for $r > 0$,

$$\boxed{\begin{aligned} V_{AB}(r) &= \frac{2 g_c^2 m_\pi}{3} (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) Y(r) \\ &\quad \times \left[\kappa^2 \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B + \left(\kappa^2 + \frac{3\kappa}{r} + \frac{3}{r^2} \right) S_{12} \right], \\ S_{12} &= 3(\boldsymbol{\sigma}_A \cdot \hat{\boldsymbol{r}})(\boldsymbol{\sigma}_B \cdot \hat{\boldsymbol{r}}) - \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B. \end{aligned}} \tag{6.12}$$

The central and tensor terms arise from the same endpoints and propagation. For total isospin $I = 0, 1$, the inner product $\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B$ takes the values $-3, +1$, respectively; for total spin $S = 0, 1$, the spin inner product likewise takes $-3, +1$. The central term is therefore attractive in channels $(I, S) = (1, 0), (0, 1)$ and repulsive in channels $(0, 0), (1, 1)$. The tensor term further distinguishes spin directions relative to the line joining the nucleons and acts on the corresponding orbital components. Identical nucleons also obey fermionic antisymmetry.

This interaction connects complete nucleon endpoints, meson propagation, and returns in both directions into a single process. The finite kernel gives the spatial response of the constructed process, while its long-wavelength form reveals the common origin of finite range, central forces, and tensor forces. Short-range multimeson interactions and nuclear bound-state spectra require the corresponding outputs to be included in the same joint space.

§7 Conclusions

A baryon persists as a whole through its collective mode and the sustained interaction that preserves it, while its internal color, flavor, and spin structure determines its transformation properties and exchange responses. In this construction, a coherently formed nucleon continues to participate in propagation and return. Its phase accumulation, survival share, and internal responses are determined by the complete joint evolution. Figure 7.1 connects this ongoing process to exchange between nucleons.

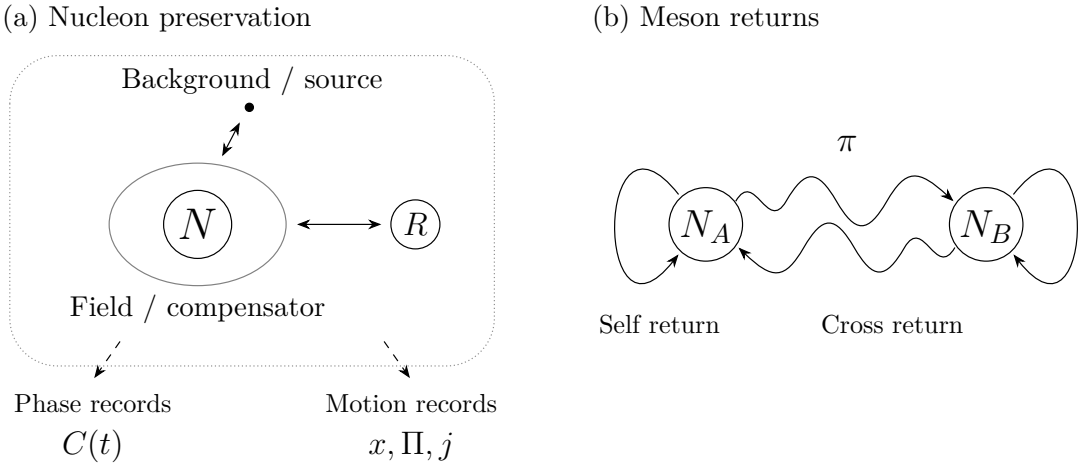


Figure 7.1. Nucleon preservation, readout, and meson returns. (a) The source, nucleon, field, compensating matter and schematic reference R evolve jointly; auxiliaries, work, prior outputs and source backreaction are retained. Dashed arrows denote separate readout settings. $C(t)$ is reference coherence; its mass interpretation requires the rest preparation specified in the text. (b) Both nucleons remain present. Return to the emitting endpoint shifts its phase; returns between different endpoints in both directions produce central and tensor terms.

Collective preservation connects spectral weights and the background interaction to a common mass scale, while the complete ports of a common exchange source determine the relative strengths of the conjugate mass arms:

$$\boxed{\begin{aligned} M_B &= E_0\sqrt{\frac{Z_L}{C_{\mathrm{bg}}}}\log Z_L, \\ \frac{\Delta m_{\mathrm{arm}}}{\mu_0} &= 2\chi_{\mathrm{arm}}\cot\alpha\tan\theta. \end{aligned}} \tag{7.1}$$

The first expression uses the specified spectral preparation and common background calibration, giving 936.669 MeV for the full-spectrum reference in Eq. (4.4). The strength in the second expression is determined by the actually supplied, simultaneously protected source ports and their routing. The common collective mode sets the overall scale, while different internal matrix elements determine differences in response. The internal flavor–spin structure also appears in magnetic responses and weak transitions.

With the field and mobile compensating matter included, phase and motion involve their joint evolution. The actual reference coherence retains all frequency

differences and complex residues of the complete H/B histories. The same mass arm enters the memory kernel for motion through propagation between orientations. Within a fixed reference branch, Eq. (4.19) gives

$$\boxed{\begin{aligned}\Sigma(t,s) = g_{\mathrm{tr}}(t) g_{\mathrm{tr}}(s) D_{A,\phi(t)} U_B(t,s) \\ \times \exp\left(i \int_s^t M(v)\, dv \right) D_{A,\phi(s)}.\end{aligned}} \tag{7.2}$$

Here $M(v)$ is the mass-arm coefficient, and U_B retains propagation of the field and compensating matter; the negative-orientation prehistory remains part of the complete evolution. Nucleon phase accumulation and transport response thus share the same internal interactions and field memory. Constraints on these quantities from coherence and continuity readouts can be propagated into bounds on motion that was not used in parameter estimation.

The same internal structure also determines the endpoints of nucleon–meson exchange. Self returns correct individual phases, while cross returns through spatial mediator propagation produce central and tensor terms of finite range.

Determining the rest mass of the complete nucleon with its field requires a pattern-preserving preparation whose phase relative to a common reference identifies the relevant spectral combination. The inertial role of this combination must then be established within the same motion kernel to compare the complete proton and neutron rest masses.

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